E-Commerce Sales Analytics

Power BI Assignment 2

Screenshot-by-screenshot steps and corrected observations

Calculated columns • DAX measures • KPI cards • Charts • Matrix • Treemap • Map • Slicers

Project objective: Build an interactive Power BI report to analyze sales, targets, profit, order counts, monthly trends, and geographic distribution.

Accuracy note: Observations are matched to the visible work in each of the 29 source screenshots. The Sales target table shown in the screenshots has Category, Month of Order Date, and Target; no Segment field is visible. The Minimum Target card therefore represents the overall minimum target.

Workflow overview

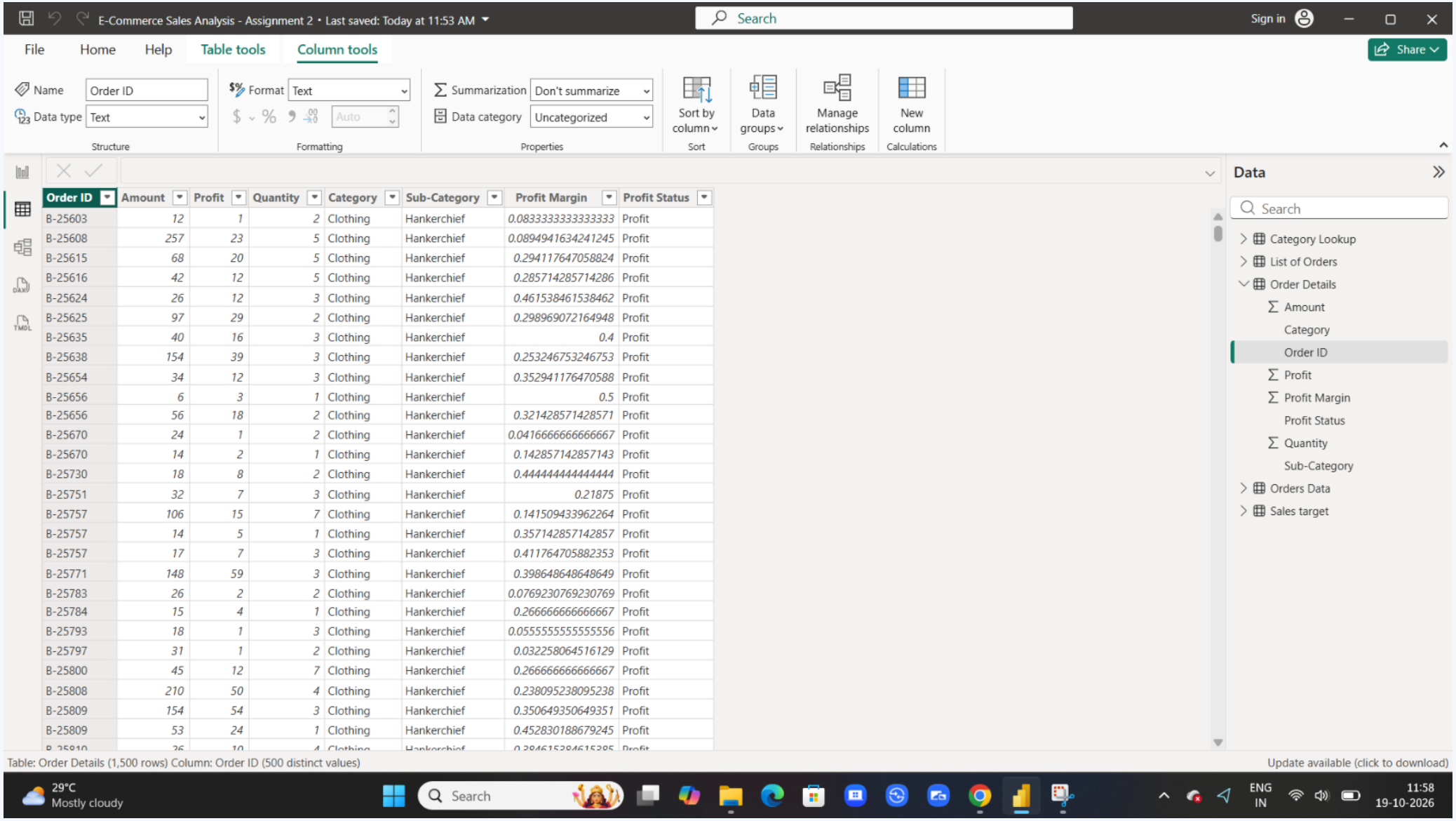
Each following page contains one original screenshot, the matching step, and an observation describing the visible work.

Screenshots	Work covered
1–6	Review source fields; create calculated columns and DAX measures.
7–11	Create and position KPI cards; style the report header; add YTD Sales.
12–17	Create target, profit-margin, monthly trend, and scatter visuals.
18–25	Build the matrix, treemap, order-count visual, and geographic map.
26–29	Add target cards, create the minimum-target measure, format the card, and review the final dashboard.

How to use this PDF on GitHub: Upload it to the same repository as your Power BI project file and reference it from the README as project documentation.

Step 1: Review the Order Details data

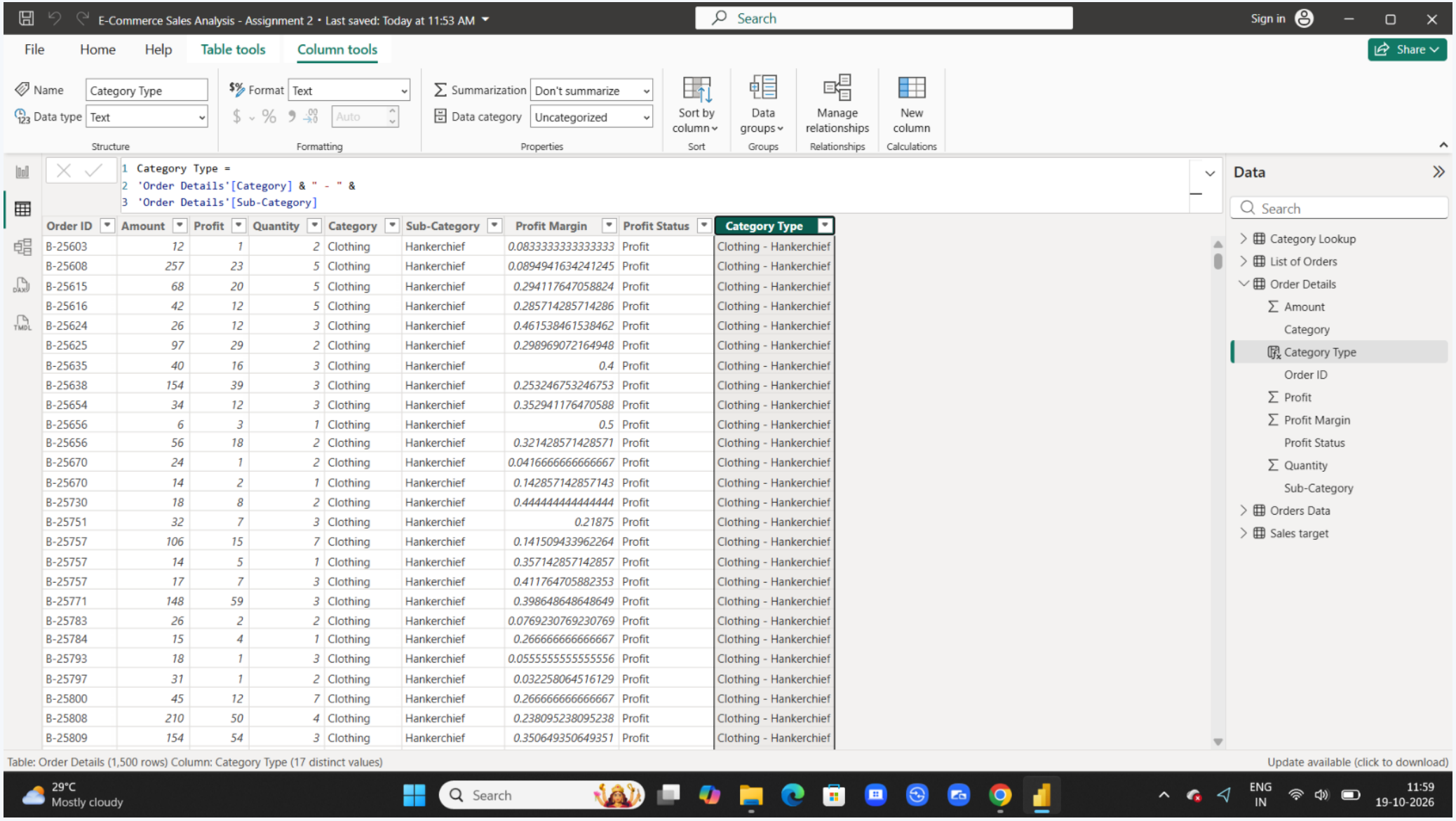
Observation: Viewed the Order Details table in Data view and inspected existing fields such as Order ID, Amount, Profit, Quantity, Category, Sub-Category, Profit Margin, and Profit Status before adding calculated columns.



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25603	12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit
B-25608	257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit
B-25615	68	20	5	Clothing	Hankerchief	0.294117647058824	Profit
B-25616	42	12	5	Clothing	Hankerchief	0.285714285714286	Profit
B-25624	26	12	3	Clothing	Hankerchief	0.461538461538462	Profit
B-25625	97	29	2	Clothing	Hankerchief	0.298969072164948	Profit
B-25635	40	16	3	Clothing	Hankerchief	0.4	Profit
B-25638	154	39	3	Clothing	Hankerchief	0.253246753246753	Profit
B-25654	34	12	3	Clothing	Hankerchief	0.352941176470588	Profit
B-25656	6	3	1	Clothing	Hankerchief	0.5	Profit
B-25656	56	18	2	Clothing	Hankerchief	0.321428571428571	Profit
B-25670	24	1	2	Clothing	Hankerchief	0.0416666666666667	Profit
B-25670	14	2	1	Clothing	Hankerchief	0.142857142857143	Profit
B-25730	18	8	2	Clothing	Hankerchief	0.444444444444444	Profit
B-25751	32	7	3	Clothing	Hankerchief	0.21875	Profit
B-25757	106	15	7	Clothing	Hankerchief	0.141509433962264	Profit
B-25757	14	5	1	Clothing	Hankerchief	0.357142857142857	Profit
B-25757	17	7	3	Clothing	Hankerchief	0.411764705882353	Profit
B-25771	148	59	3	Clothing	Hankerchief	0.398648648648649	Profit
B-25783	26	2	2	Clothing	Hankerchief	0.0769230769230769	Profit
B-25784	15	4	1	Clothing	Hankerchief	0.266666666666667	Profit
B-25793	18	1	3	Clothing	Hankerchief	0.0555555555555556	Profit
B-25797	31	1	2	Clothing	Hankerchief	0.032258064516129	Profit
B-25800	45	12	7	Clothing	Hankerchief	0.266666666666667	Profit
B-25808	210	50	4	Clothing	Hankerchief	0.238095238095238	Profit
B-25809	154	54	3	Clothing	Hankerchief	0.350649350649351	Profit
B-25809	53	24	1	Clothing	Hankerchief	0.452830188679245	Profit

Step 2: Create the Category Type calculated column

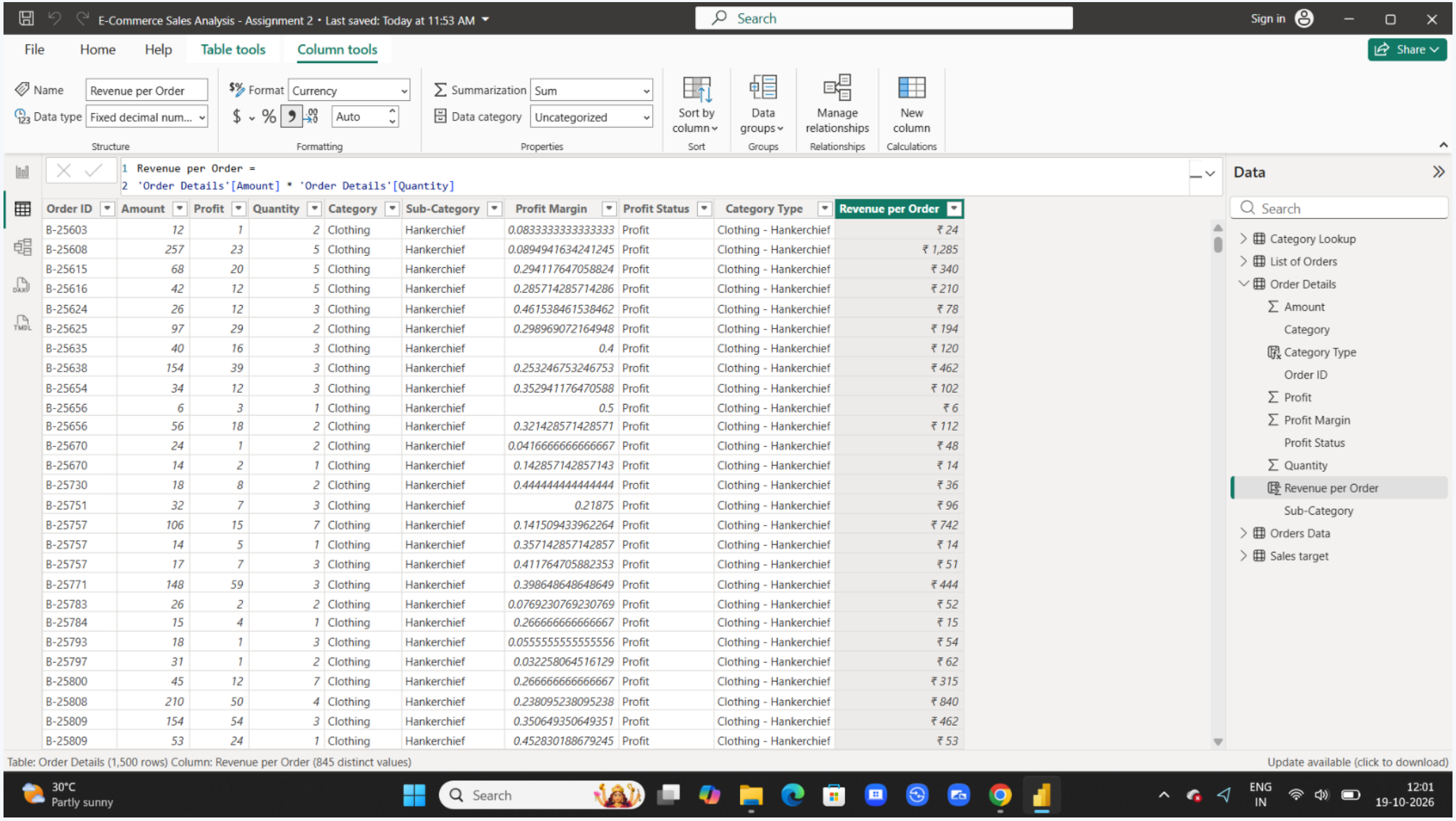
Observation: Entered a DAX expression that combines Category and Sub-Category into one text value, then checked the new Category Type column in the table.



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type
B-25603	12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing - Hankerchief
B-25608	257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing - Hankerchief
B-25615	68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing - Hankerchief
B-25616	42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing - Hankerchief
B-25624	26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing - Hankerchief
B-25625	97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing - Hankerchief
B-25635	40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing - Hankerchief
B-25638	154	39	3	Clothing	Hankerchief	0.253246753246753	Profit	Clothing - Hankerchief
B-25654	34	12	3	Clothing	Hankerchief	0.352941176470588	Profit	Clothing - Hankerchief
B-25656	6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing - Hankerchief
B-25656	56	18	2	Clothing	Hankerchief	0.321428571428571	Profit	Clothing - Hankerchief
B-25670	24	1	2	Clothing	Hankerchief	0.0416666666666667	Profit	Clothing - Hankerchief
B-25670	14	2	1	Clothing	Hankerchief	0.142857142857143	Profit	Clothing - Hankerchief
B-25730	18	8	2	Clothing	Hankerchief	0.444444444444444	Profit	Clothing - Hankerchief
B-25751	32	7	3	Clothing	Hankerchief	0.21875	Profit	Clothing - Hankerchief
B-25757	106	15	7	Clothing	Hankerchief	0.141509433962264	Profit	Clothing - Hankerchief
B-25757	14	5	1	Clothing	Hankerchief	0.357142857142857	Profit	Clothing - Hankerchief
B-25757	17	7	3	Clothing	Hankerchief	0.411764705882353	Profit	Clothing - Hankerchief
B-25771	148	59	3	Clothing	Hankerchief	0.398648648648649	Profit	Clothing - Hankerchief
B-25783	26	2	2	Clothing	Hankerchief	0.0769230769230769	Profit	Clothing - Hankerchief
B-25784	15	4	1	Clothing	Hankerchief	0.266666666666667	Profit	Clothing - Hankerchief
B-25793	18	1	3	Clothing	Hankerchief	0.0555555555555556	Profit	Clothing - Hankerchief
B-25797	31	1	2	Clothing	Hankerchief	0.032258064516129	Profit	Clothing - Hankerchief
B-25800	45	12	7	Clothing	Hankerchief	0.266666666666667	Profit	Clothing - Hankerchief
B-25808	210	50	4	Clothing	Hankerchief	0.238095238095238	Profit	Clothing - Hankerchief
B-25809	154	54	3	Clothing	Hankerchief	0.350649350649351	Profit	Clothing - Hankerchief

Step 3: Create the Revenue per Order calculated column

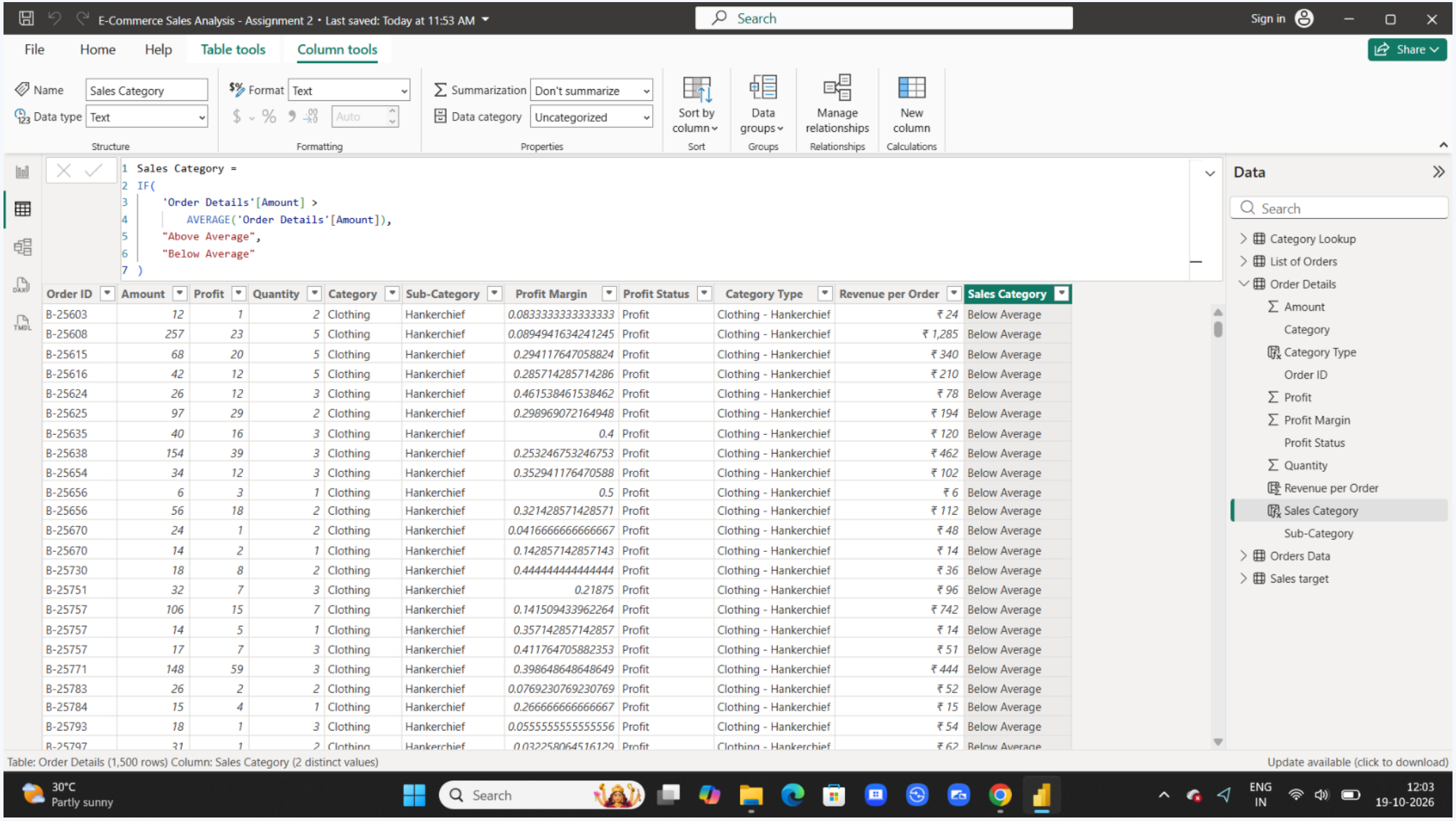
Observation: Created Revenue per Order by multiplying Amount by Quantity. The new column appears in the table with calculated values.



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue per Order
B-25603	12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing - Hankerchief	₹ 24
B-25608	257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing - Hankerchief	₹ 1,285
B-25615	68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing - Hankerchief	₹ 340
B-25616	42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing - Hankerchief	₹ 210
B-25624	26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing - Hankerchief	₹ 78
B-25625	97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing - Hankerchief	₹ 194
B-25635	40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing - Hankerchief	₹ 120
B-25638	154	39	3	Clothing	Hankerchief	0.253246753246753	Profit	Clothing - Hankerchief	₹ 462
B-25654	34	12	3	Clothing	Hankerchief	0.352941176470588	Profit	Clothing - Hankerchief	₹ 102
B-25656	6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing - Hankerchief	₹ 6
B-25656	56	18	2	Clothing	Hankerchief	0.321428571428571	Profit	Clothing - Hankerchief	₹ 112
B-25670	24	1	2	Clothing	Hankerchief	0.0416666666666667	Profit	Clothing - Hankerchief	₹ 48
B-25670	14	2	1	Clothing	Hankerchief	0.142857142857143	Profit	Clothing - Hankerchief	₹ 14
B-25730	18	8	2	Clothing	Hankerchief	0.444444444444444	Profit	Clothing - Hankerchief	₹ 36
B-25751	32	7	3	Clothing	Hankerchief	0.21875	Profit	Clothing - Hankerchief	₹ 96
B-25757	106	15	7	Clothing	Hankerchief	0.141509433962264	Profit	Clothing - Hankerchief	₹ 742
B-25757	14	5	1	Clothing	Hankerchief	0.357142857142857	Profit	Clothing - Hankerchief	₹ 14
B-25757	17	7	3	Clothing	Hankerchief	0.411764705882353	Profit	Clothing - Hankerchief	₹ 51
B-25771	148	59	3	Clothing	Hankerchief	0.398648648648649	Profit	Clothing - Hankerchief	₹ 444
B-25783	26	2	2	Clothing	Hankerchief	0.0769230769230769	Profit	Clothing - Hankerchief	₹ 52
B-25784	15	4	1	Clothing	Hankerchief	0.266666666666667	Profit	Clothing - Hankerchief	₹ 15
B-25793	18	1	3	Clothing	Hankerchief	0.0555555555555556	Profit	Clothing - Hankerchief	₹ 54
B-25797	31	1	2	Clothing	Hankerchief	0.032258064516129	Profit	Clothing - Hankerchief	₹ 62
B-25800	45	12	7	Clothing	Hankerchief	0.266666666666667	Profit	Clothing - Hankerchief	₹ 315
B-25808	210	50	4	Clothing	Hankerchief	0.238095238095238	Profit	Clothing - Hankerchief	₹ 840
B-25809	154	54	3	Clothing	Hankerchief	0.350649350649351	Profit	Clothing - Hankerchief	₹ 462
B-25809	53	24	1	Clothing	Hankerchief	0.452830188679245	Profit	Clothing - Hankerchief	₹ 53

Step 4: Create the Sales Category calculated column

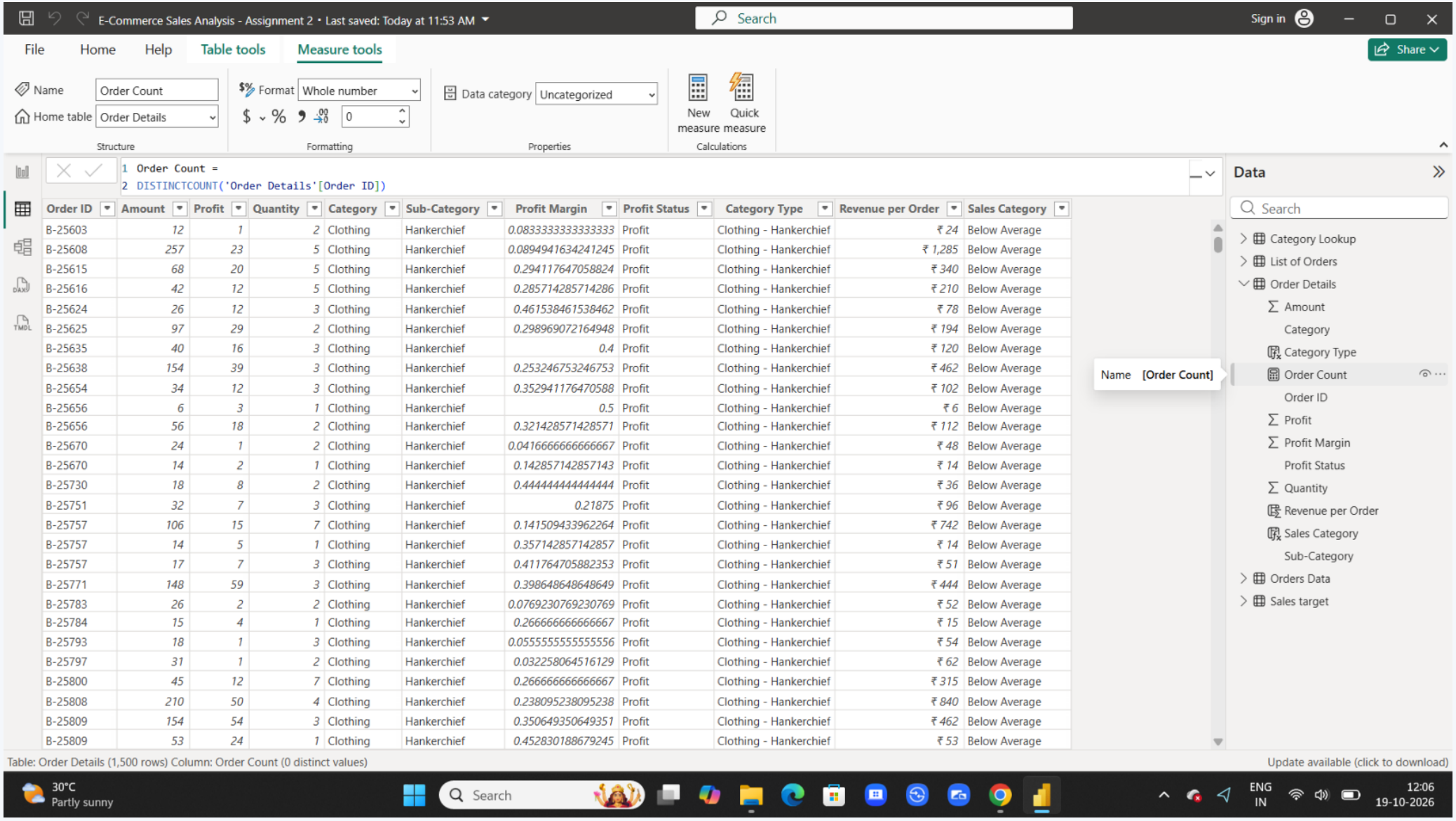
Observation: Created Sales Category with IF logic that compares each Amount with the average Amount and labels the row Above Average or Below Average.



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue per Order	Sales Category
B-25603	12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing - Hankerchief	₹ 24	Below Average
B-25608	257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing - Hankerchief	₹ 1,285	Below Average
B-25615	68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing - Hankerchief	₹ 340	Below Average
B-25616	42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing - Hankerchief	₹ 210	Below Average
B-25624	26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing - Hankerchief	₹ 78	Below Average
B-25625	97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing - Hankerchief	₹ 194	Below Average
B-25635	40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing - Hankerchief	₹ 120	Below Average
B-25638	154	39	3	Clothing	Hankerchief	0.253246753246753	Profit	Clothing - Hankerchief	₹ 462	Below Average
B-25654	34	12	3	Clothing	Hankerchief	0.352941176470588	Profit	Clothing - Hankerchief	₹ 102	Below Average
B-25656	6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing - Hankerchief	₹ 6	Below Average
B-25656	56	18	2	Clothing	Hankerchief	0.321428571428571	Profit	Clothing - Hankerchief	₹ 112	Below Average
B-25670	24	1	2	Clothing	Hankerchief	0.0416666666666667	Profit	Clothing - Hankerchief	₹ 48	Below Average
B-25670	14	2	1	Clothing	Hankerchief	0.142857142857143	Profit	Clothing - Hankerchief	₹ 14	Below Average
B-25730	18	8	2	Clothing	Hankerchief	0.444444444444444	Profit	Clothing - Hankerchief	₹ 36	Below Average
B-25751	32	7	3	Clothing	Hankerchief	0.21875	Profit	Clothing - Hankerchief	₹ 96	Below Average
B-25757	106	15	7	Clothing	Hankerchief	0.141509433962264	Profit	Clothing - Hankerchief	₹ 742	Below Average
B-25757	14	5	1	Clothing	Hankerchief	0.357142857142857	Profit	Clothing - Hankerchief	₹ 14	Below Average
B-25757	17	7	3	Clothing	Hankerchief	0.411764705882353	Profit	Clothing - Hankerchief	₹ 51	Below Average
B-25771	148	59	3	Clothing	Hankerchief	0.398648648648649	Profit	Clothing - Hankerchief	₹ 444	Below Average
B-25783	26	2	2	Clothing	Hankerchief	0.0769230769230769	Profit	Clothing - Hankerchief	₹ 52	Below Average
B-25784	15	4	1	Clothing	Hankerchief	0.266666666666667	Profit	Clothing - Hankerchief	₹ 15	Below Average
B-25793	18	1	3	Clothing	Hankerchief	0.0555555555555556	Profit	Clothing - Hankerchief	₹ 54	Below Average
B-25797	31	1	2	Clothing	Hankerchief	0.032258064516129	Profit	Clothing - Hankerchief	₹ 62	Below Average

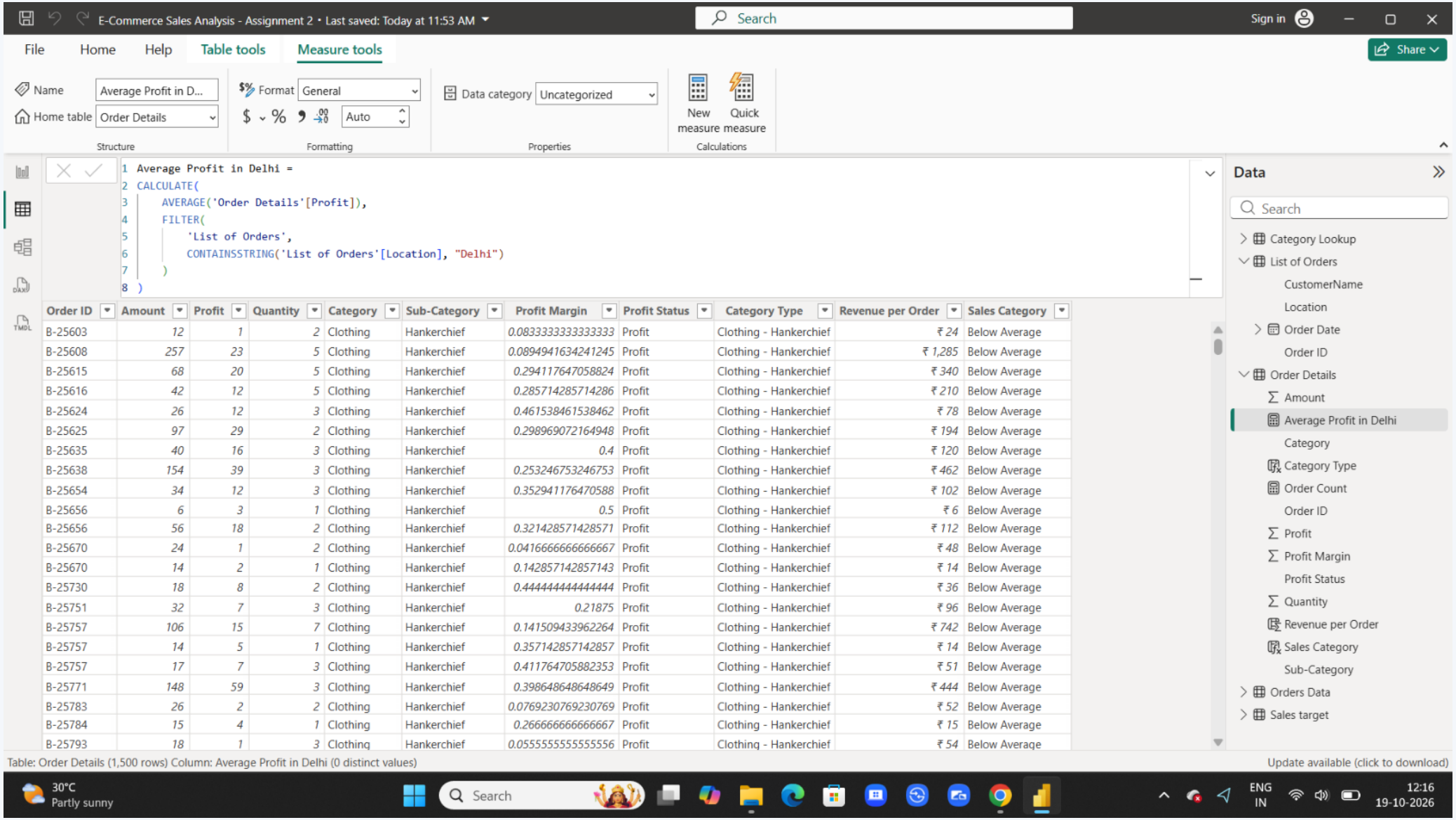
Step 5: Create the Order Count measure

Observation: Created the Order Count measure using DISTINCTCOUNT on Order ID to count unique orders rather than counting every order-detail row.



Step 6: Create the Average Profit in Delhi measure

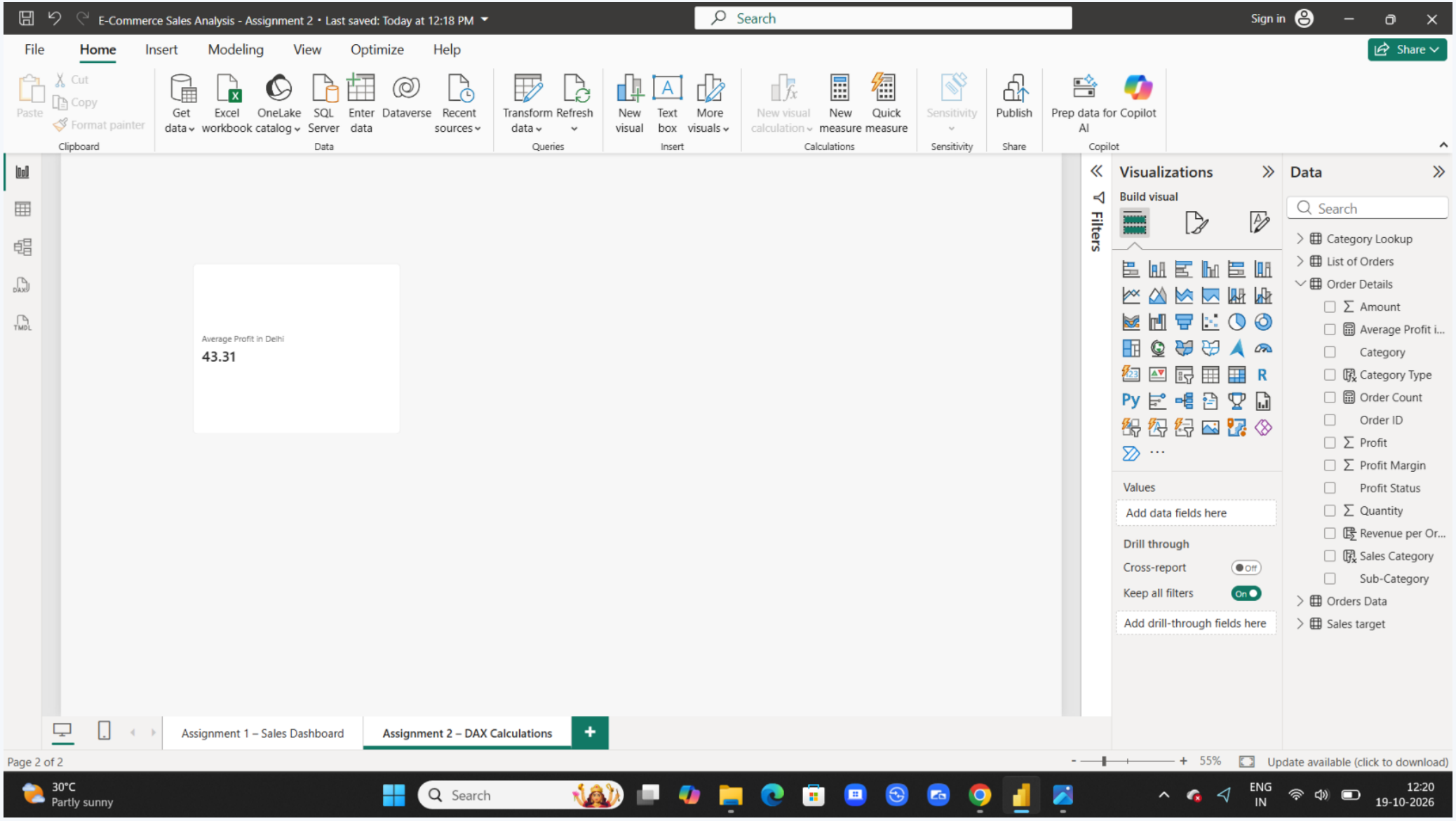
Observation: Entered the Average Profit in Delhi measure using CALCULATE, AVERAGE of Profit, and a location filter that identifies Delhi in the List of Orders table.



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue per Order	Sales Category
B-25603	12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing - Hankerchief	₹ 24	Below Average
B-25608	257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing - Hankerchief	₹ 1,285	Below Average
B-25615	68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing - Hankerchief	₹ 340	Below Average
B-25616	42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing - Hankerchief	₹ 210	Below Average
B-25624	26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing - Hankerchief	₹ 78	Below Average
B-25625	97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing - Hankerchief	₹ 194	Below Average
B-25635	40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing - Hankerchief	₹ 120	Below Average
B-25638	154	39	3	Clothing	Hankerchief	0.253246753246753	Profit	Clothing - Hankerchief	₹ 462	Below Average
B-25654	34	12	3	Clothing	Hankerchief	0.352941176470588	Profit	Clothing - Hankerchief	₹ 102	Below Average
B-25656	6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing - Hankerchief	₹ 6	Below Average
B-25656	56	18	2	Clothing	Hankerchief	0.321428571428571	Profit	Clothing - Hankerchief	₹ 112	Below Average
B-25670	24	1	2	Clothing	Hankerchief	0.0416666666666667	Profit	Clothing - Hankerchief	₹ 48	Below Average
B-25670	14	2	1	Clothing	Hankerchief	0.142857142857143	Profit	Clothing - Hankerchief	₹ 14	Below Average
B-25730	18	8	2	Clothing	Hankerchief	0.444444444444444	Profit	Clothing - Hankerchief	₹ 36	Below Average
B-25751	32	7	3	Clothing	Hankerchief	0.21875	Profit	Clothing - Hankerchief	₹ 96	Below Average
B-25757	106	15	7	Clothing	Hankerchief	0.141509433962264	Profit	Clothing - Hankerchief	₹ 742	Below Average
B-25757	14	5	1	Clothing	Hankerchief	0.357142857142857	Profit	Clothing - Hankerchief	₹ 14	Below Average
B-25757	17	7	3	Clothing	Hankerchief	0.411764705882353	Profit	Clothing - Hankerchief	₹ 51	Below Average
B-25771	148	59	3	Clothing	Hankerchief	0.398648648648649	Profit	Clothing - Hankerchief	₹ 444	Below Average
B-25783	26	2	2	Clothing	Hankerchief	0.0769230769230769	Profit	Clothing - Hankerchief	₹ 52	Below Average
B-25784	15	4	1	Clothing	Hankerchief	0.266666666666667	Profit	Clothing - Hankerchief	₹ 15	Below Average
B-25793	18	1	3	Clothing	Hankerchief	0.0555555555555556	Profit	Clothing - Hankerchief	₹ 54	Below Average

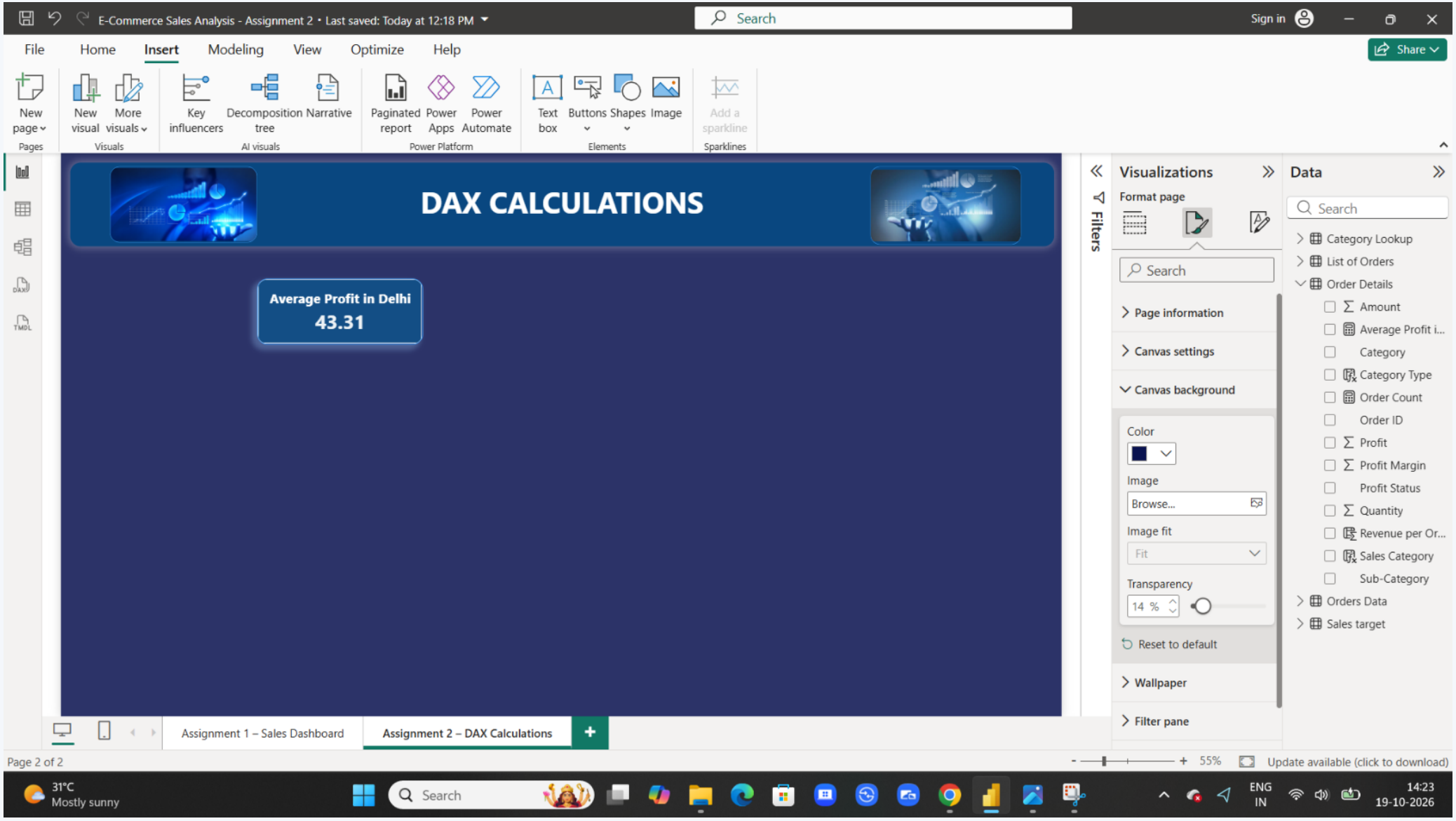
Step 7: Add a Card for Average Profit in Delhi

Observation: Inserted a Card visual on the Assignment 2 report page and assigned the Average Profit in Delhi measure to its value. At this stage, the card was still on a mostly blank canvas.



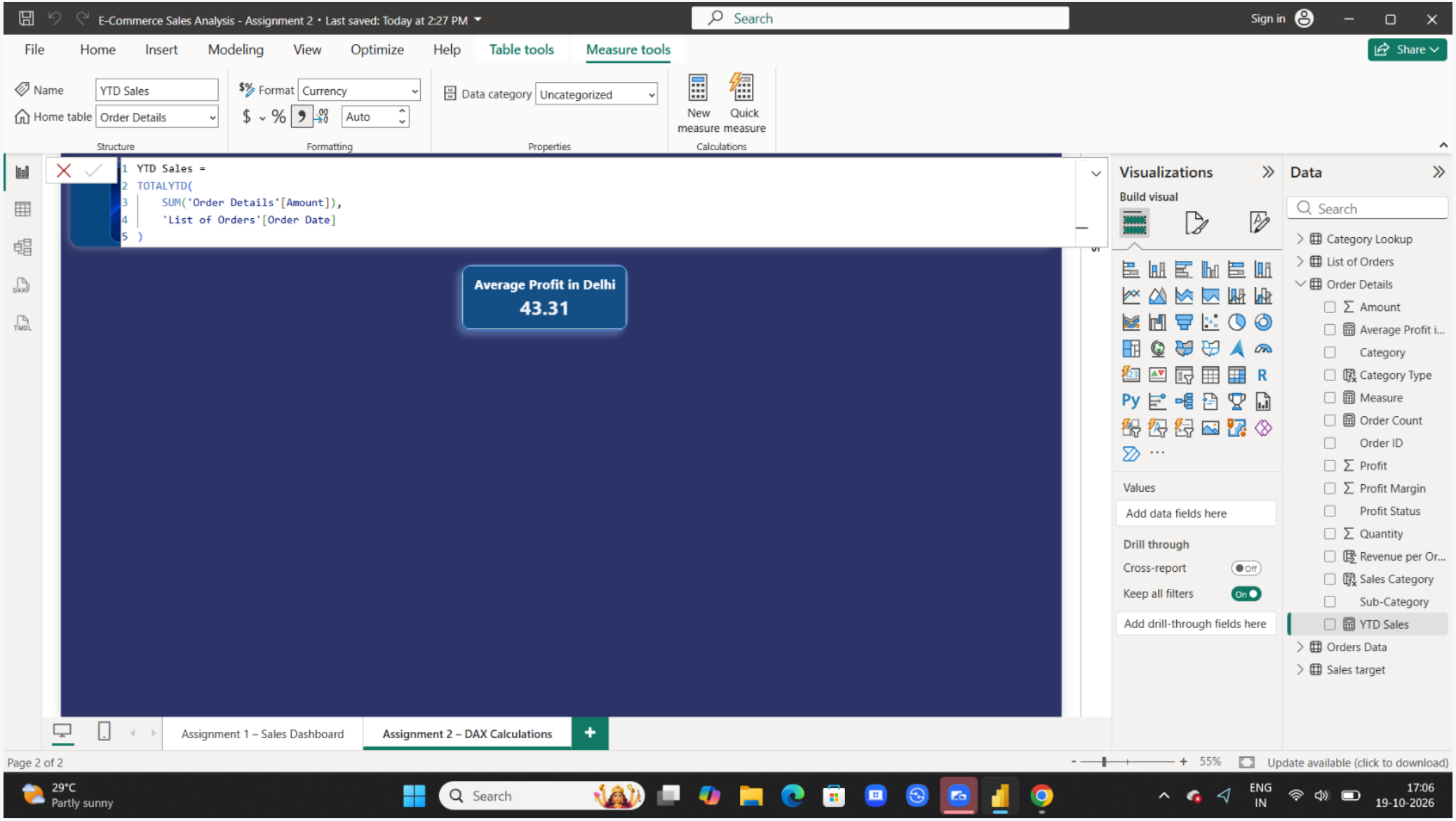
Step 8: Design the report page header

Observation: Formatted the report-page background and added the blue header area with decorative images and the title DAX CALCULATIONS.



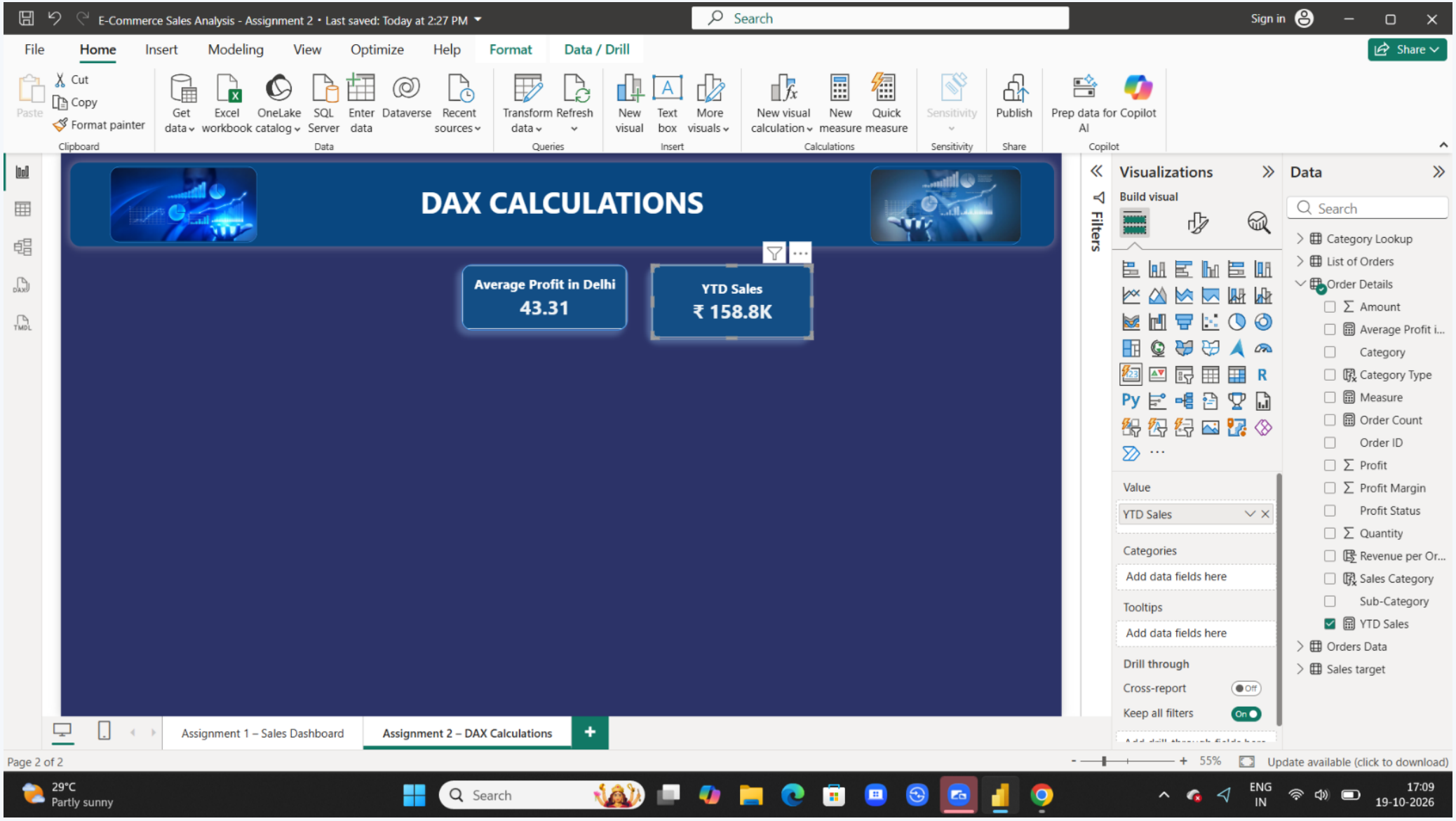
Step 9: Create the YTD Sales measure

Observation: Entered the YTD Sales DAX measure using TOTALYTD, SUM of Order Details[Amount], and List of Orders[Order Date]. The screenshot shows the measure being written in the formula bar.



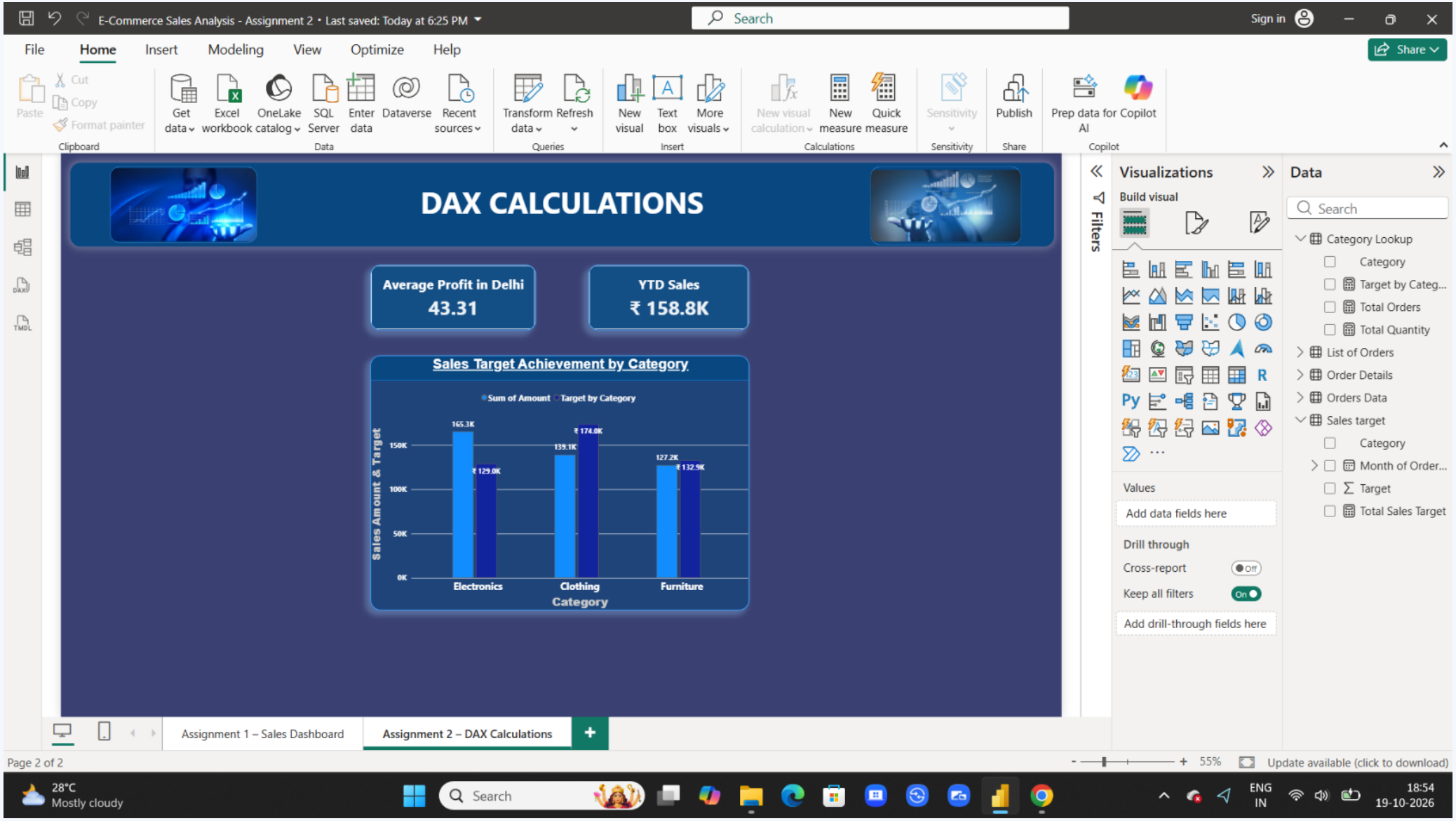
Step 10: Add the YTD Sales card

Observation: Added the YTD Sales measure to a Card visual. The card displays approximately ■158.8K next to the Average Profit in Delhi card.



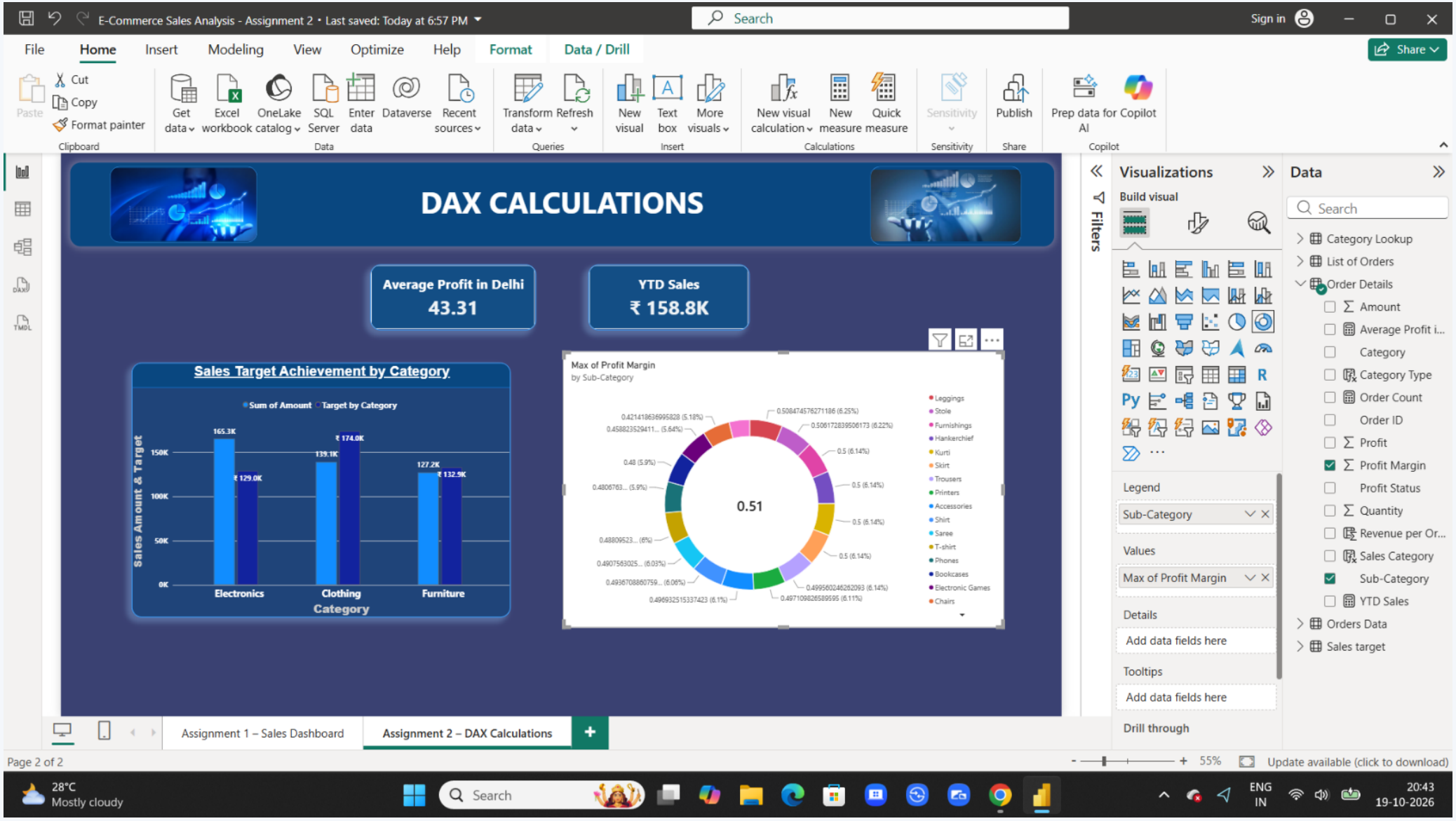
Step 11: Create the Sales Target Achievement by Category chart

Observation: Created a clustered column chart comparing Sum of Amount with Target by Category. Electronics, Clothing, and Furniture are shown so sales can be compared with their targets.



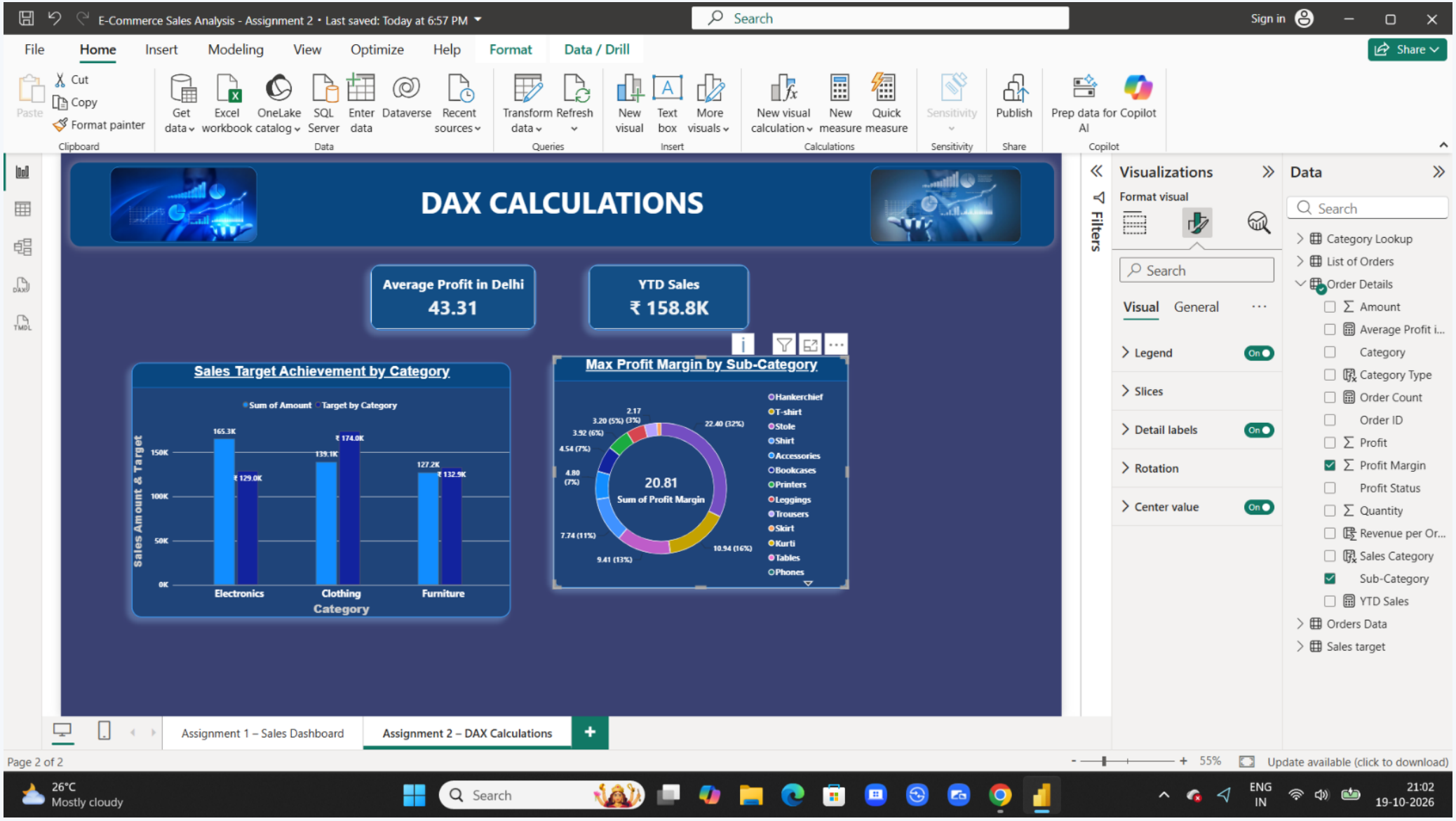
Step 12: Create the Max Profit Margin by Sub-Category chart

Observation: Added a donut chart using Sub-Category as the legend and the maximum Profit Margin as the value.



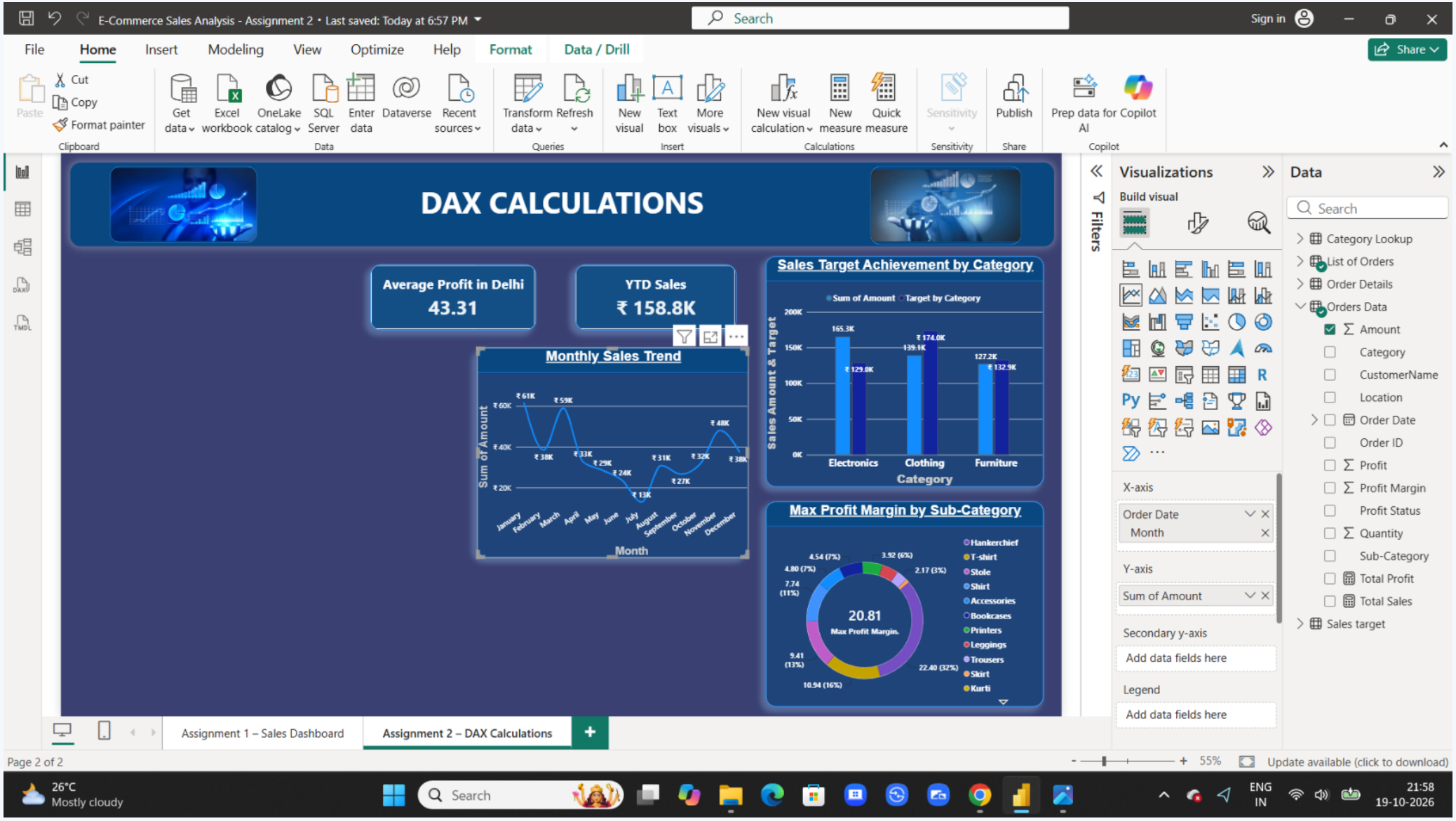
Step 13: Format the Max Profit Margin chart

Observation: Adjusted the donut chart's layout and title, then positioned it alongside the other report visuals.



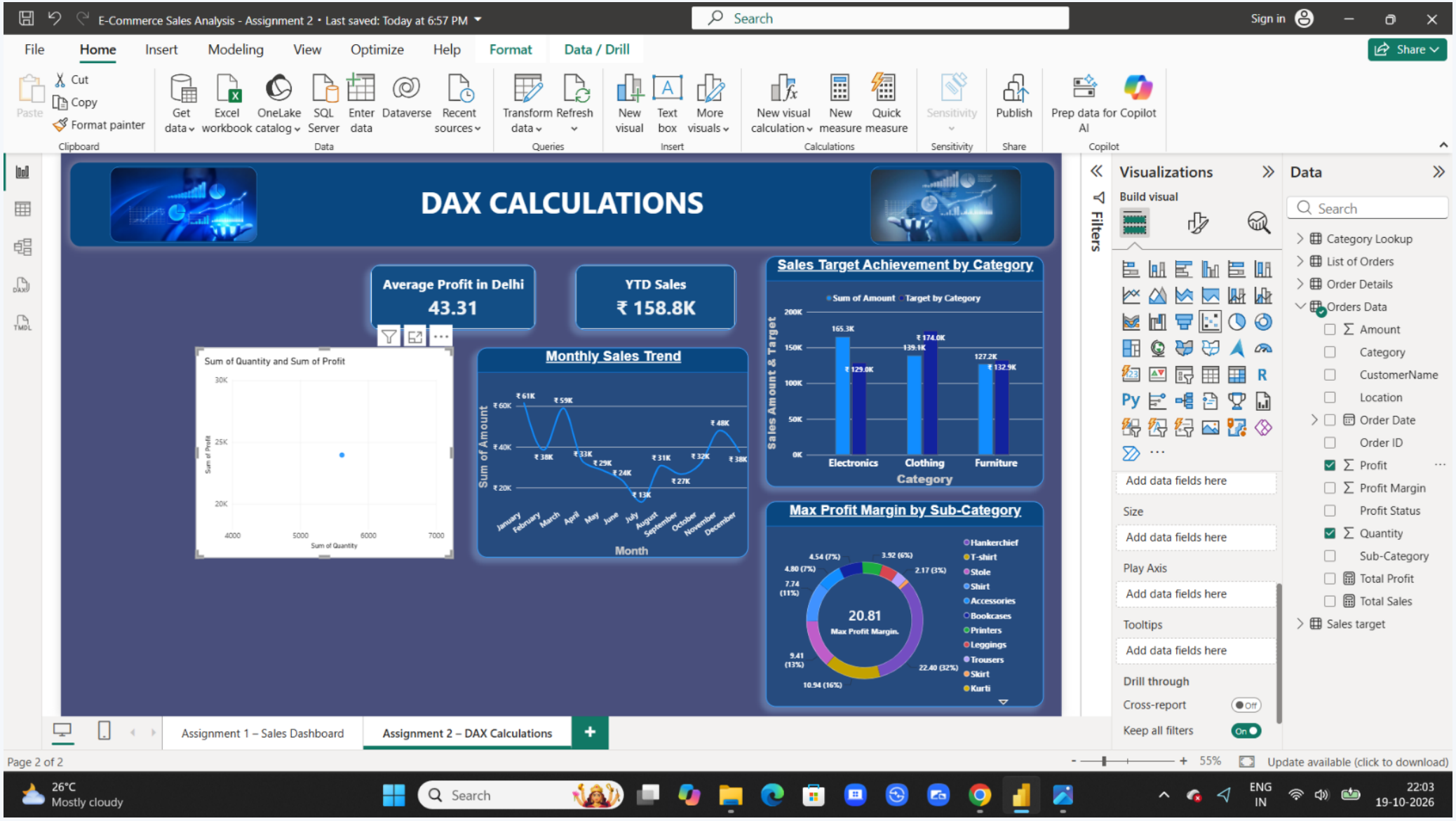
Step 14: Create the Monthly Sales Trend chart

Observation: Added a line chart with Month on the horizontal axis and Sum of Amount on the vertical axis to show monthly sales changes.



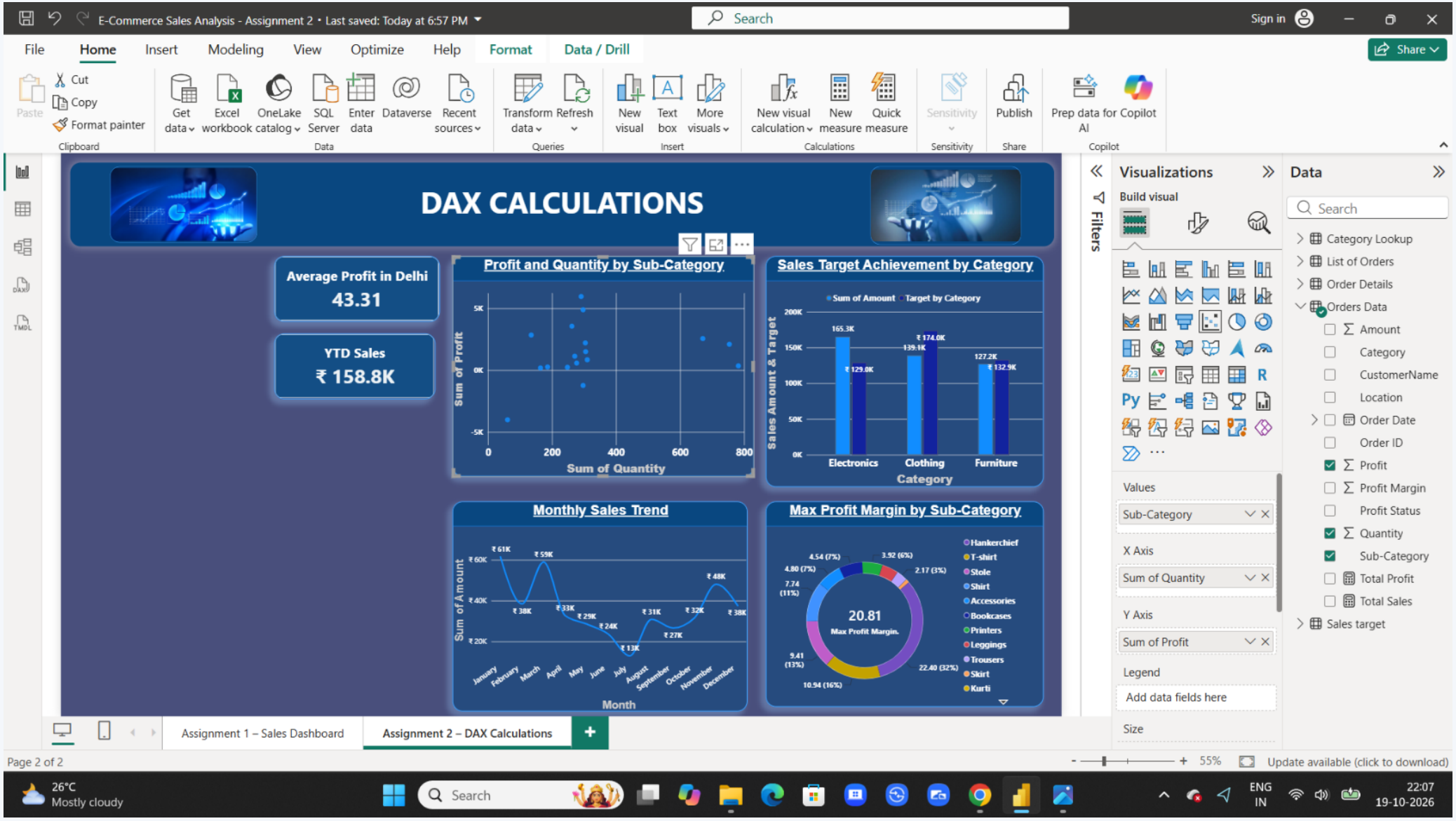
Step 15: Create the Profit and Quantity Analysis chart

Observation: Added a scatter chart to compare Quantity and Profit across the available product or sub-category groupings.



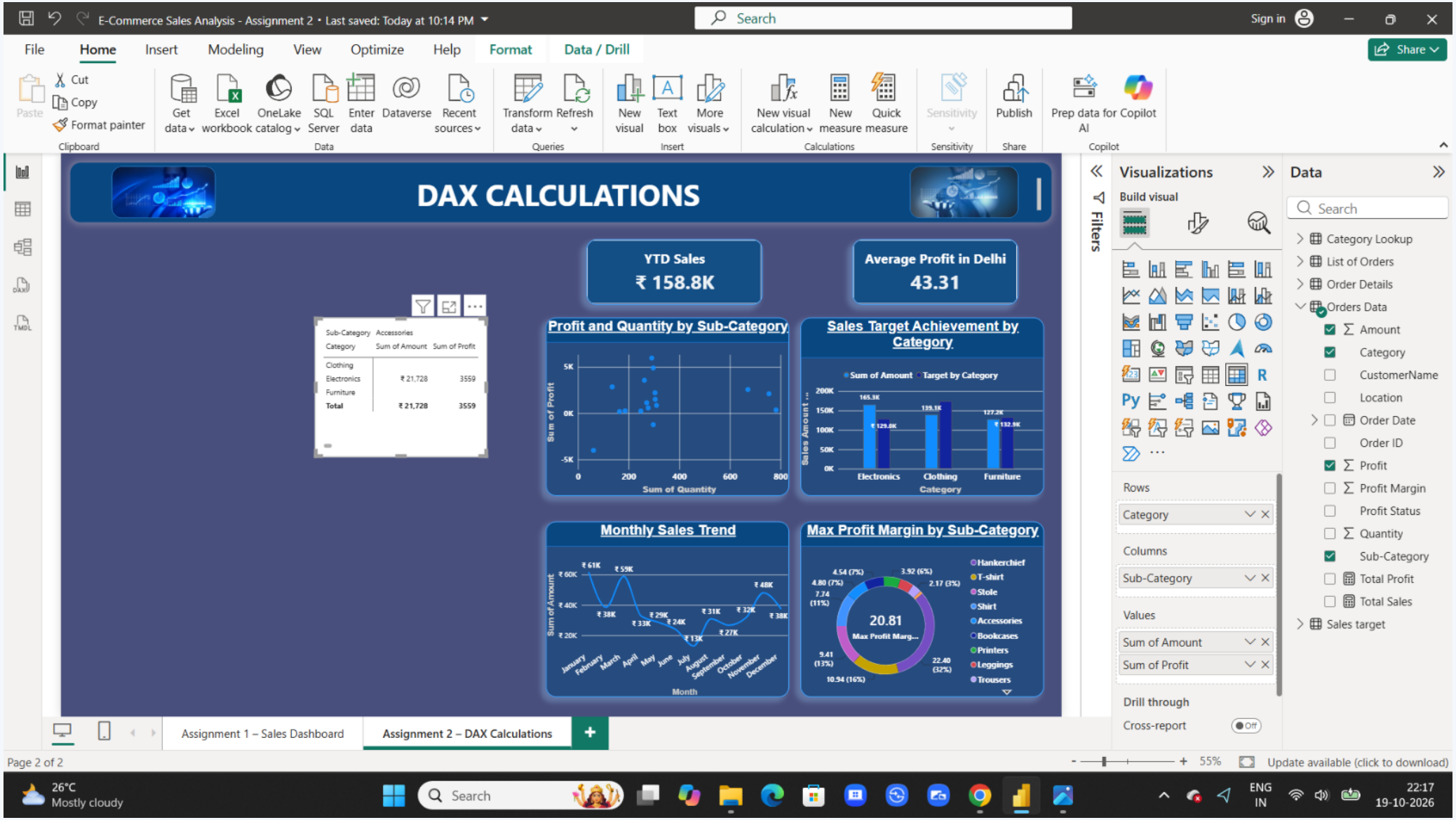
Step 16: Arrange the main analytical charts

Observation: Resized and positioned the target comparison, donut, monthly trend, and scatter visuals to begin organizing the report page.



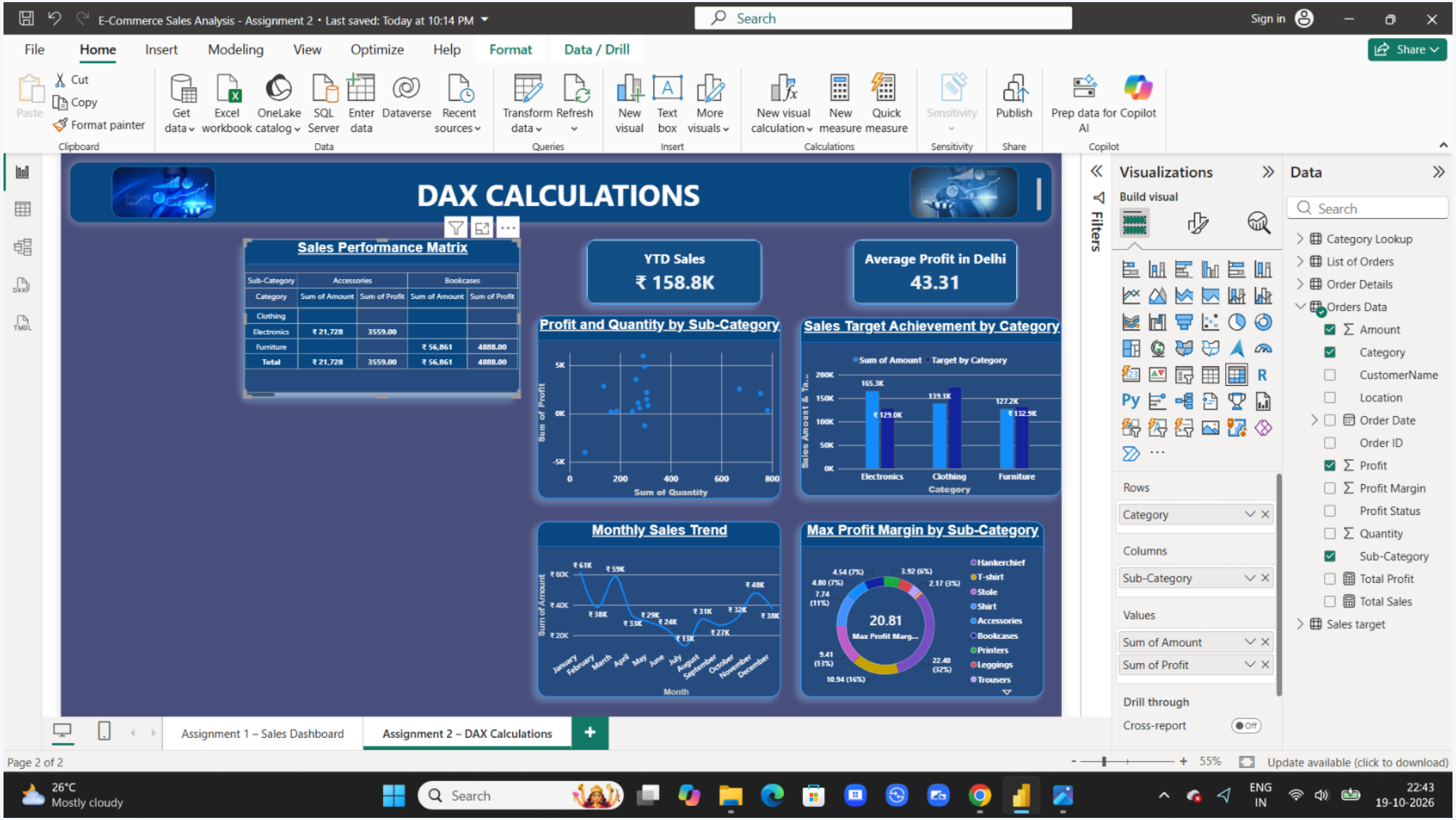
Step 17: Create the Sales Performance Matrix

Observation: Added a matrix with Category as rows, Sub-Category as columns, and Sum of Amount and Sum of Profit as values for comparison.



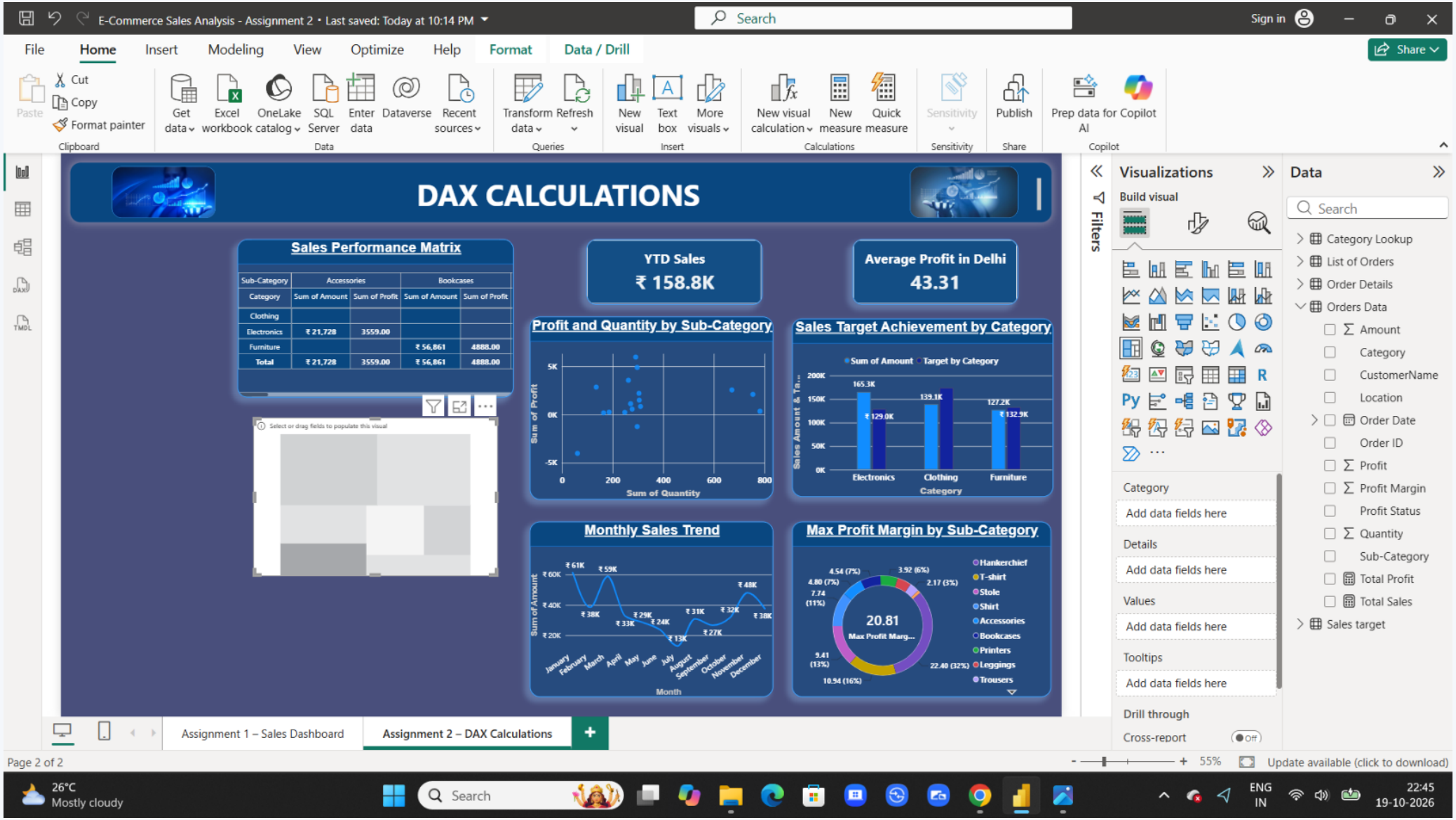
Step 18: Position the Sales Performance Matrix

Observation: Positioned the matrix in the report layout alongside the existing charts.



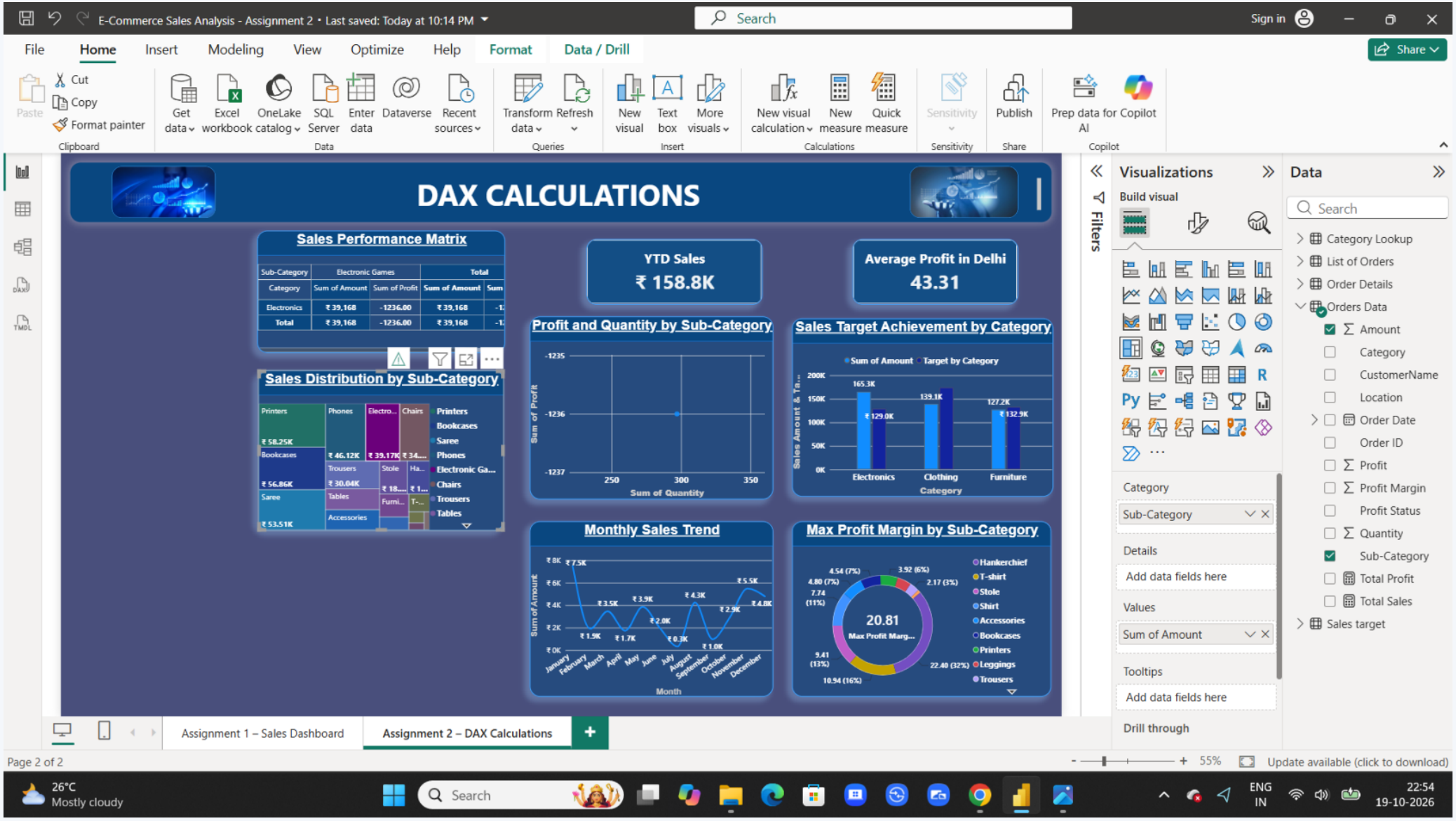
Step 19: Add the Sales Distribution treemap

Observation: Inserted a treemap to show sales Amount distribution across Category and Sub-Category.



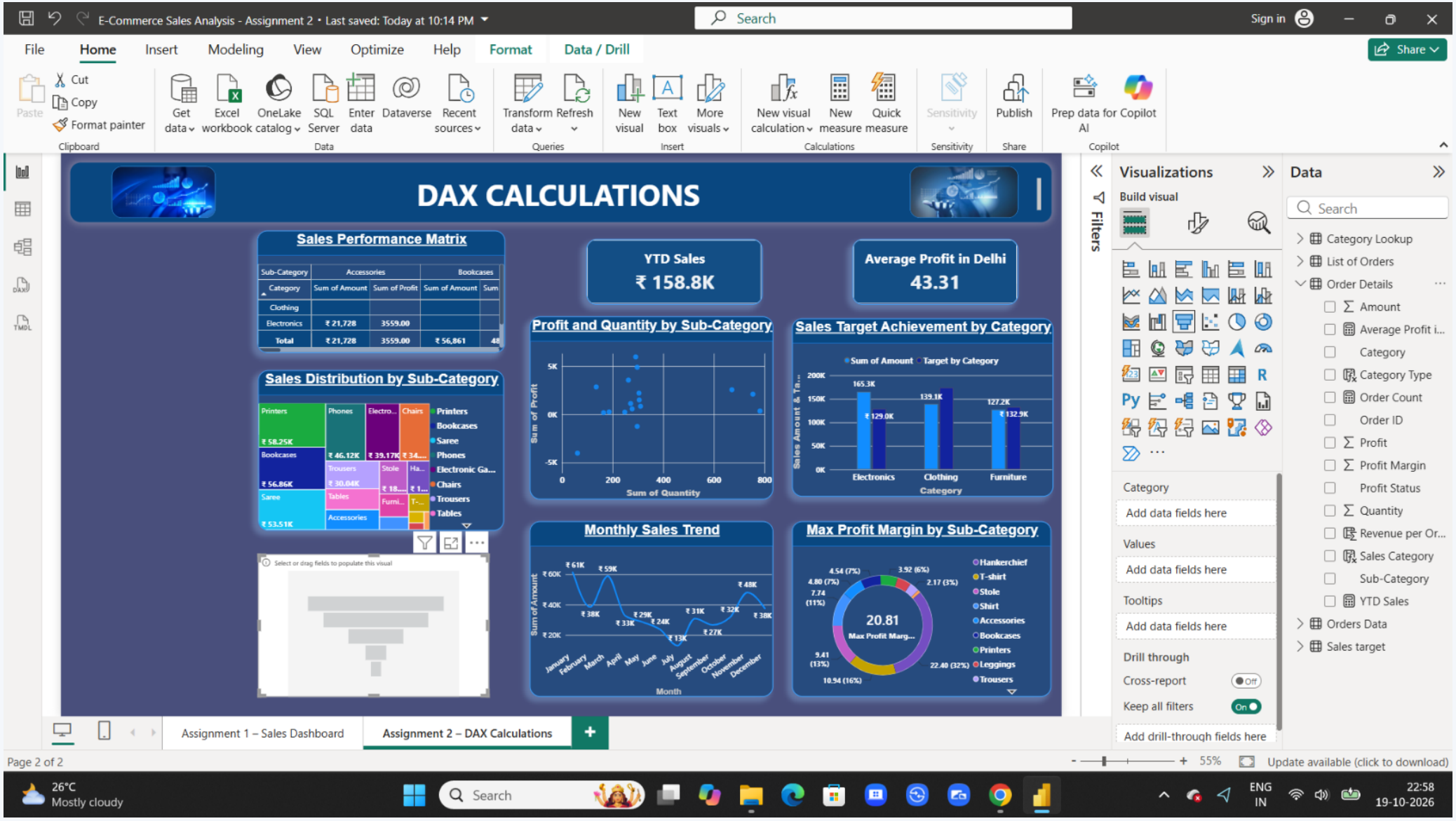
Step 20: Complete and position the treemap

Observation: Configured the treemap to display sales distribution by sub-category and positioned it with the other dashboard visuals.



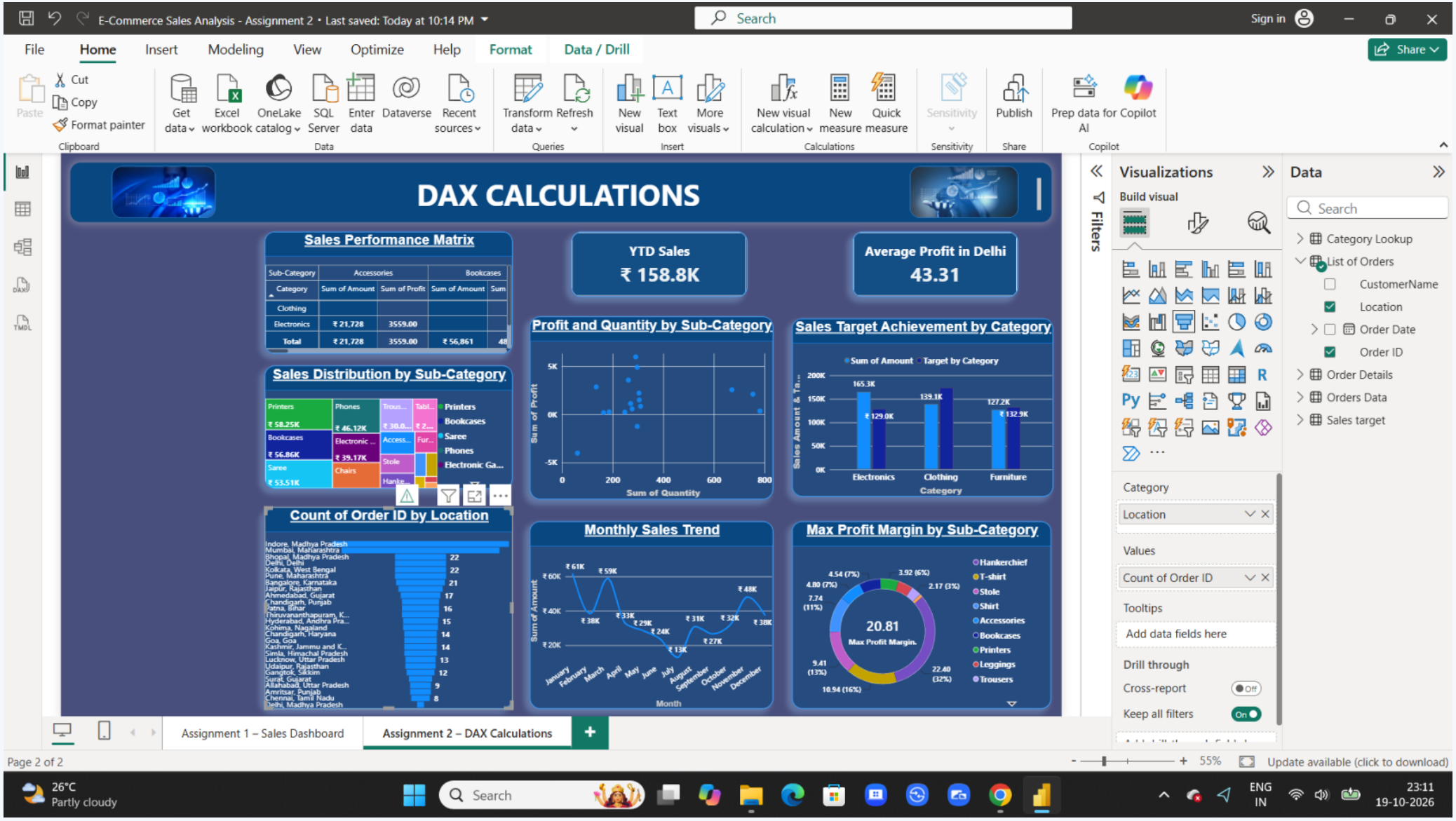
Step 21: Create the Order Count Analysis visual

Observation: Added a funnel visual using Location as the category and the order-count field as the value to compare counts across locations.



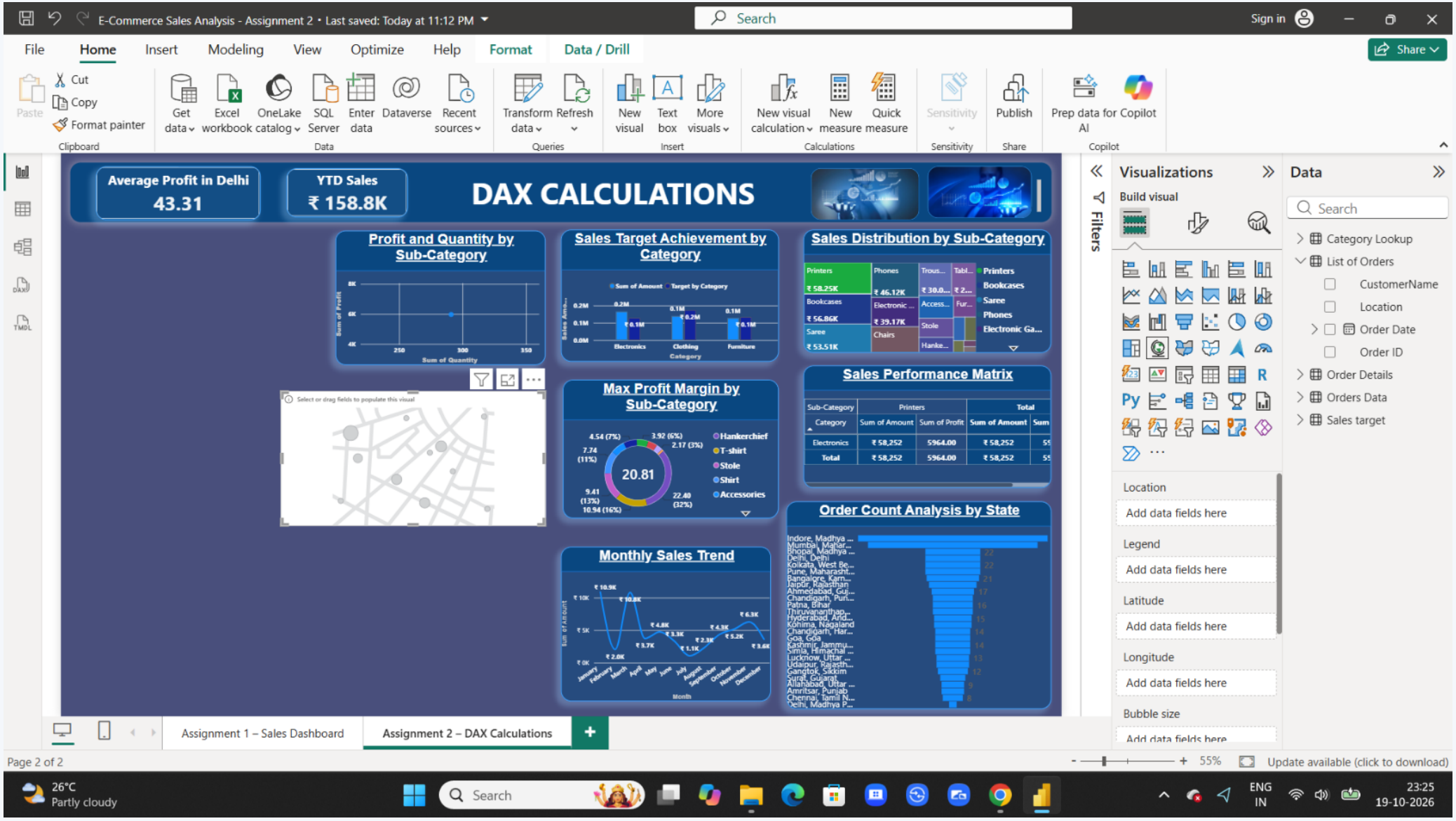
Step 22: Position the Order Count Analysis visual

Observation: Placed the order-count visual on the report page and checked that location names and counts were displayed.



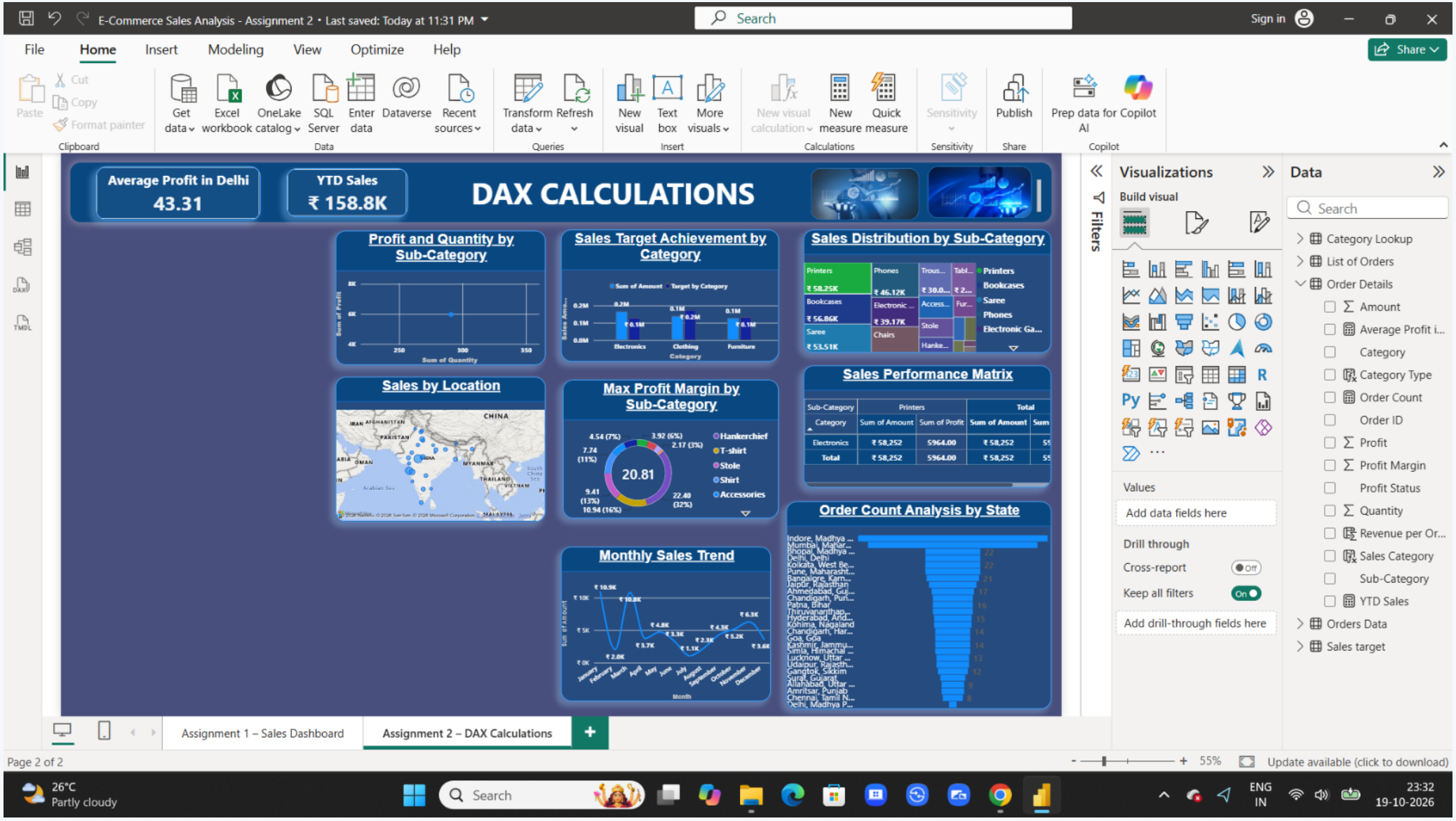
Step 23: Configure the Sales by Location map

Observation: Added and configured a map visual using location information and sales Amount to explore geographic sales.



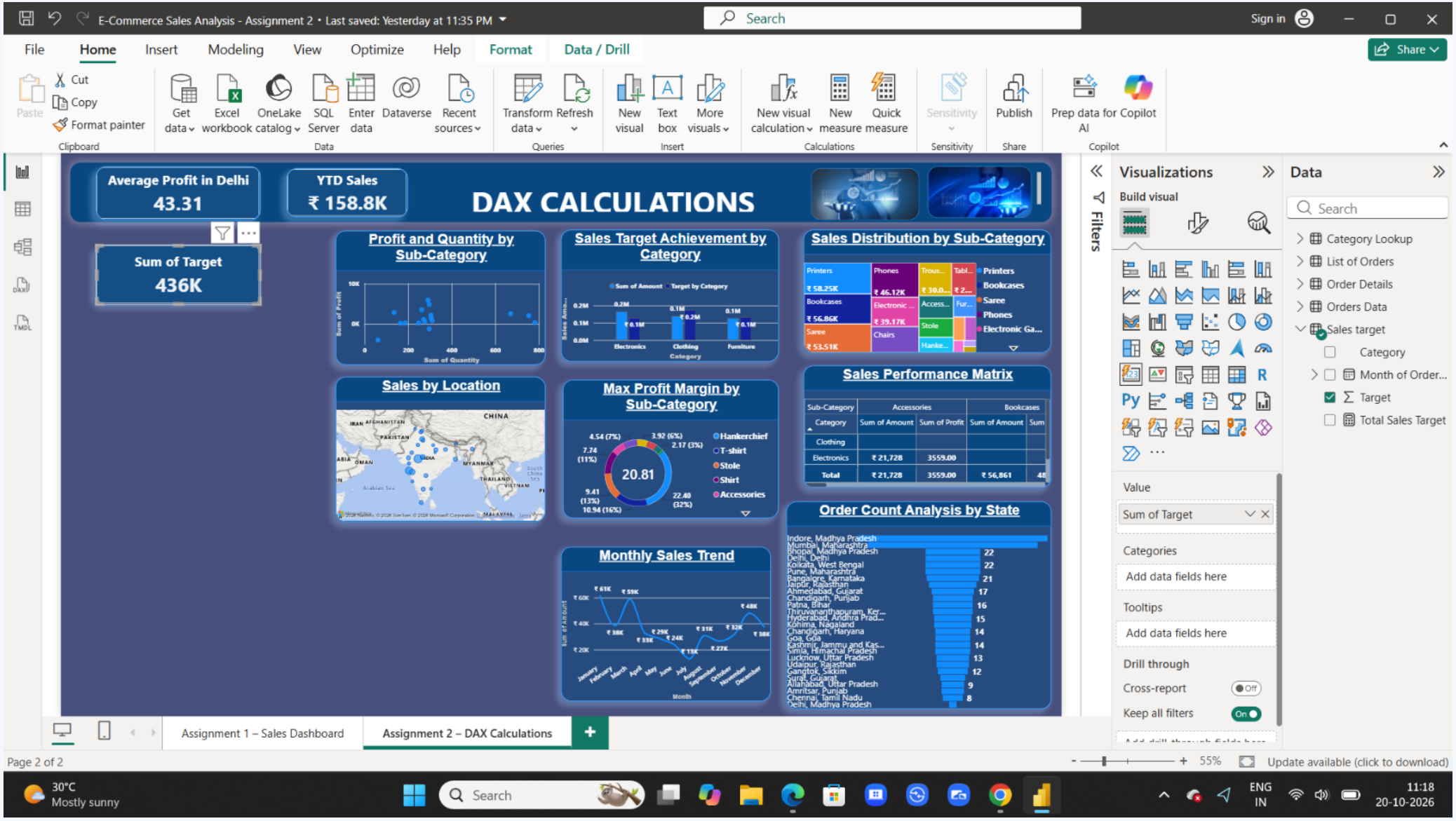
Step 24: Display and position the Sales by Location map

Observation: Placed the geographic sales map with the other dashboard visuals. The map displays bubbles for locations in the dataset.



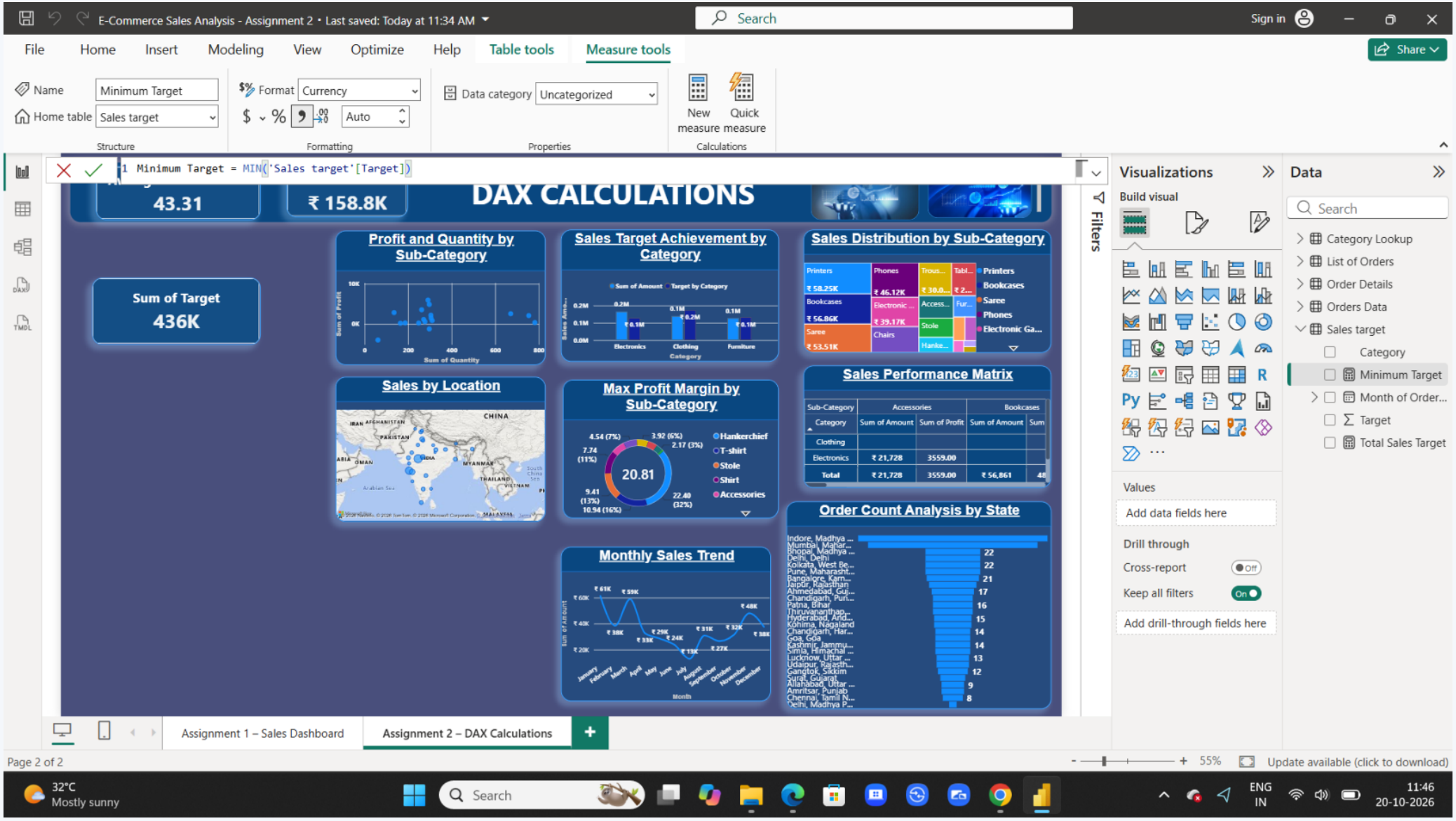
Step 25: Add the Sum of Target card

Observation: Added a Card visual using the target total. The card displays approximately 436K as the overall target.



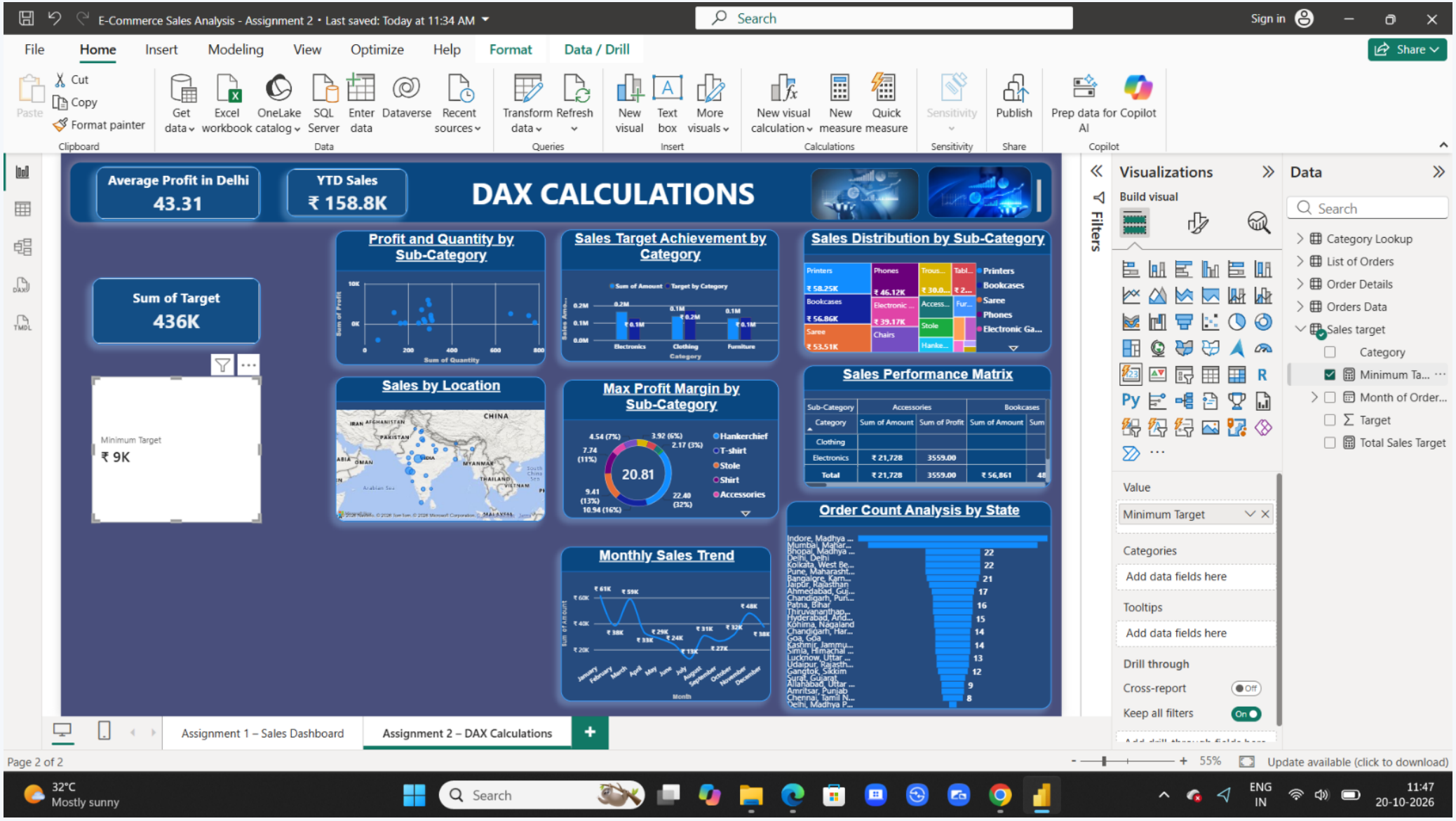
Step 26: Create the Minimum Target measure

Observation: Created the Minimum Target measure using MIN on the Target column in the Sales target table.



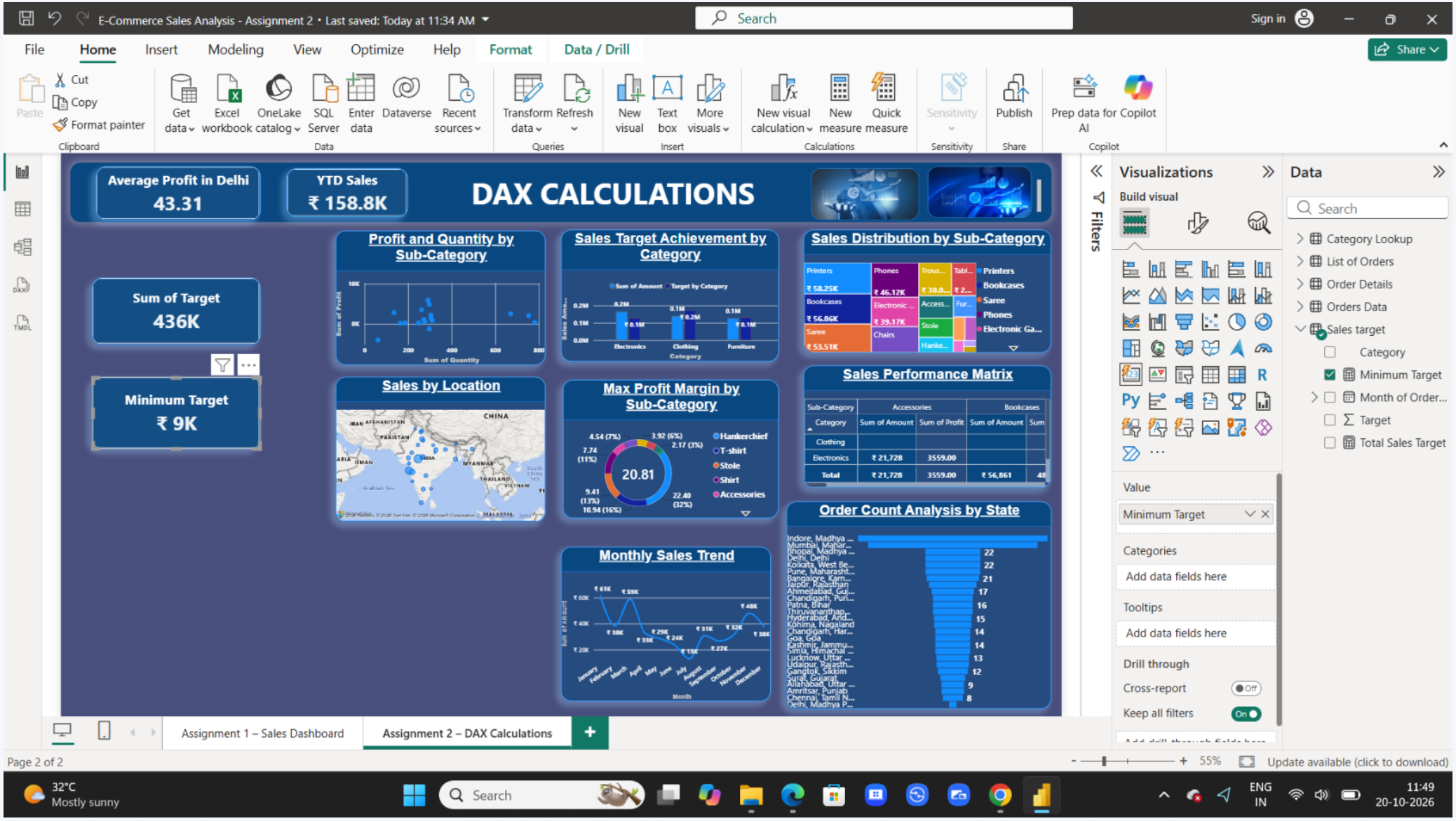
Step 27: Add the Minimum Target card

Observation: Placed the Minimum Target measure in a Card visual. The card shows the overall minimum target value.



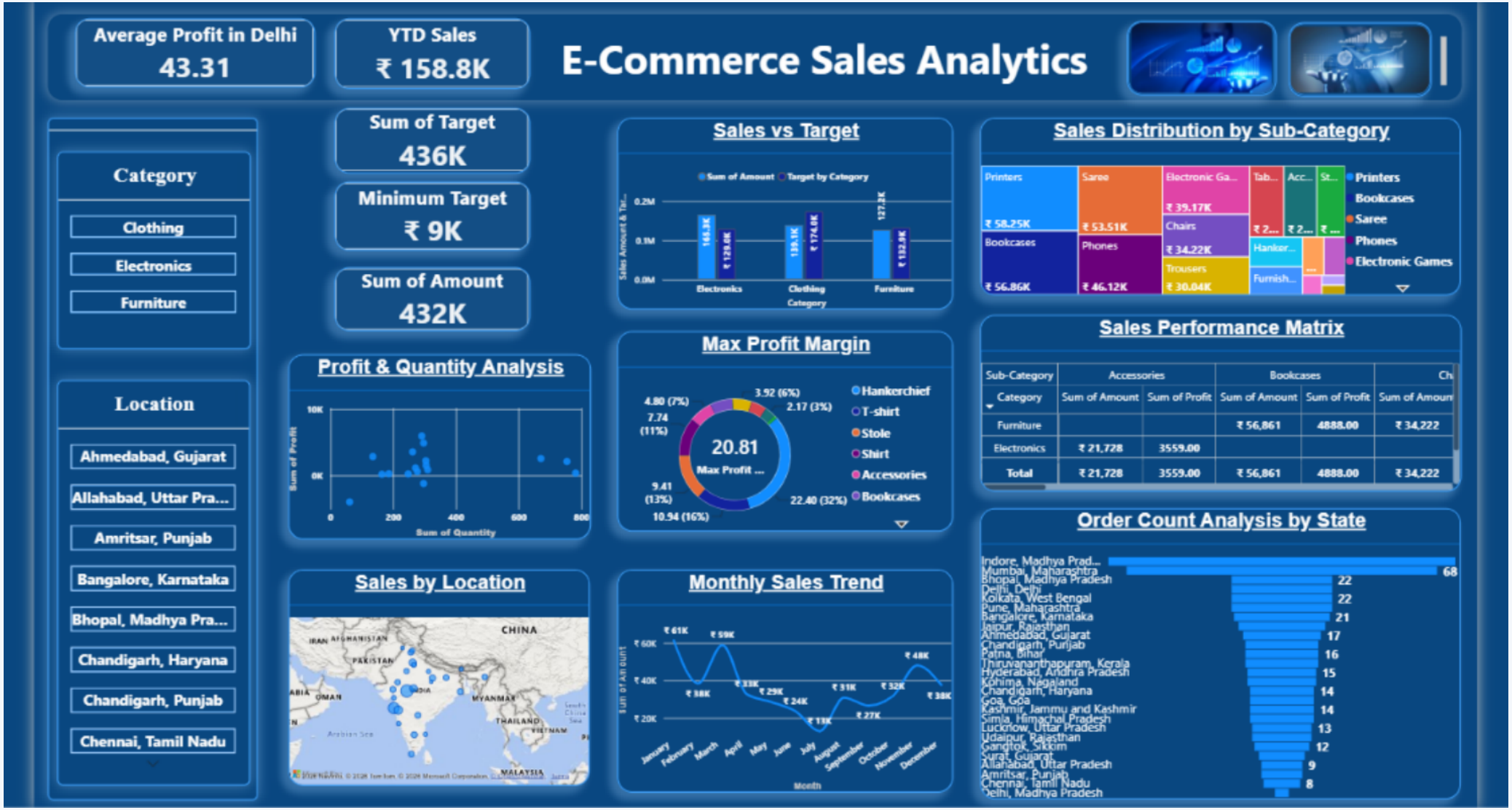
Step 28: Format the Minimum Target card

Observation: Styled and positioned the Minimum Target card to match the dark-blue dashboard design.



Step 29: Review the final dashboard

Observation: Reviewed the final dashboard with KPI cards, Category and Location slicers, target comparison, treemap, matrix, scatter chart, donut chart, map, monthly trend, and order-count analysis. The Minimum Target card shows the overall minimum, not a minimum by segment, because no Segment field is visible in the provided dataset screenshots.



Final result: The completed report combines the available KPI cards, slicers, and analytical visuals on one dashboard page. Save the PBIX file and keep a backup before sharing.